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MILITARY SPECIFICATION

INDICATORS, ATTITUDE DIRECTOR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers three types of remote attitude-director indicators with superimposed flight-director information.

1.2 Classification. Indicators shall be of the following types, as specified (see 6.2):

ARU-11/A	Without horizontal pointer alarm flag
ARU-21/A	With horizontal pointer alarm flag
ARU-31/A	With precise bank presentation, without horizontal pointer alarm flag.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

QQ-P-416	Plating, Cadmium (Electrodeposited)
* PPP-B-601	Box, Wood, Cleated-Plywood
PPP-B-636	Box, Fiberboard

Military

* MIL-P-116	Preservation-Packaging, Methods of
MIL-E-5400	Electronic Equipment, Airborne, General Specification for
MIL-C-5541	Chemical Conversion Coatings on Aluminum and Aluminum Alloys
MIL-I-6181	Interference Control Requirements, Aircraft Equipment
MIL-A-8625	Anodic Coatings, for Aluminum and Aluminim Alloys

FSC 6610

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MIL-C-14806	Coating, Reflection Reducing, for Instrument Cover Glasses and Lighting Wedges
MIL-M-18012	Marking for Aircrew Station Displays Design and Configuration of
MIL-L-25467	Lighting, Integral, Aircraft Instrument, General Specification for
MIL-P-26514	Polyurethane Foam, Rigid or Plastic, for Packaging
MIL-L-27160	Lighting, Instrument, Integral, White, General Specification for
MIL-B-27497	Bearings, Jewel, Sapphire or Ruby, Synthetic
MIL-B-83292	Bearing, Ball, Annular, Instrument, for Precision Rotating Components

STANDARDS

Federal

FED-STD-595	Colors

* Military

MIL-STD-100	Engineering Drawing Practices
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of US Military Property
MIL-STD-143	Standards and Specifications, Order of Precedence for the Selection of
MIL-STD-470	Maintainability Program Requirements
MIL-STD-471	Maintainability Verification/Demonstration/Evaluation
MIL-STD-704	Electric Power, Aircraft, Characteristics and Utilization of
MIL-STD-781	Reliability Tests, Exponential Distribution
MIL-STD-794	Parts and Equipment, Procedures for Packaging and Packing of
MIL-STD-810	Environmental Test Methods
MIL-STD-831	Test Reports, Preparation of
MIL-STD-838	Lubrication of Military Equipment
MIL-STD-889	Dissimilar Metals
MIL-STD-1472	Human Engineering Design Criteria for Military Systems, Equipment and Facilities
MS17322	Meter, Time Totalizing, Miniature, Digital, 115 Volt 400 Cycle
MS25271	Relay, 10 Amp, 4 Pdt, Class B8, Sealed, with Solder Hooks
MS33545	Case - Instrument 5 x 5-1/4, Standard Dimensions for
MS33558	Numerals and Letters, Aircraft Instrument Dial, Standard Form

(Copies of documents required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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* PUBLICATIONS

Federal Aviation Administration

FAA Advisory Circular 20-69

(Request for copies should be addressed to Department of Transportation, Federal Aviation Administration, Washington, D.C. 20591.)

3. REQUIREMENTS

3.1 Qualification. Indicators furnished under this specification shall be products which are qualified for listing on the applicable qualified products list at the time set for opening of bids (see 4.4 and 6.4).

3.2 Data. Unless otherwise specified in the contract or order (see 6.2), no data other than reports and drawings accompanying qualification samples are required by this specification or any of the documents referenced in section 2.

3.3 Selection of standards and specifications. Standards and specifications for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.4 Materials

3.4.1 Fungus-proof materials. Materials that are nutrients for fungi shall not be used where it is practical to avoid them. Where used and not hermetically sealed, they shall be treated with a fungicidal agent acceptable to the qualifying activity. If used in a hermetically sealed enclosure, fungicidal treatment will not be necessary.

3.4.2 Metals. Metals shall be corrosion resistant or suitably treated to resist corrosion due to fuels, salt spray, or atmospheric conditions likely to be met in storage or normal service.

3.4.2.1 Dissimilar metals. Unless suitably protected against electrolytic corrosion, dissimilar metals as defined in MIL-STD-889 shall not be used in intimate contact with each other.

3.4.3 Nonmagnetic materials. Nonmagnetic materials shall be used for all parts except where magnetic materials are essential.

3.4.4 Corrosive fumes. Materials, as installed in the indicator and under the service conditions specified herein, shall not liberate deleterious fumes.

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3.4.5 Protective treatment. Materials used in the construction of the indicator that are subject to deterioration when exposed to climatic and environmental conditions likely to occur during service usage shall be protected against such deterioration in a manner that will in no way prevent compliance with the performance requirements of this specification. The use of any protective coating that will crack, chip, or scale with age or extremes of climatic and environmental conditions shall be avoided.

3.5 Design and construction

3.5.1 Design. The indicator shall be designed to display pitch, roll, and heading information with full freedom around each axis, fligh director information controlled by a remote computer, and turn and slip information.

3.5.1.1 Electronic components. Electronic components and the application thereof shall be in accordance with the class 2 requirements of MIL-E-5400 except as modified herein.

3.5.1.2 Maintainability

3.5.1.2.1 The indicator shall be designed using the maintainability design criteria specified in MIL-STD-470 in addition to the following requirements:

a. The design of the indicator shall be such as to allow rapid and positive recognition of malfunction or marginal performance.

b. The indicator shall be designed such that optimum accessibility exists to all components requiring inspection, removal, or replacement.

c. The indicator shall be designed such that a minimum number and types of tools and peculiar test equipment shall be needed to perform repair.

d. The indicator shall be designed to minimize the complexity of maintenance tasks required during repair. (For example: calibration, adjustments, inspection, etc.). This includes interchangeability of like components and use of standardized equipment or commercial items.

3.5.1.2.2 The indicator shall be maintained at three levels of maintenance as described below:

a. Field level. Maintenance shall consist of trouble shooting and removal and replacement of the indicator in the aircraft.

b. Intermediate level. (Field shop). Maintenance shall consist of functional testing, and external adjustments, (and replacement of amplifier power supply.)

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c. Depot level. Maintenance shall consist of complete disassembly, failure isolation, repair and replacement of internal components and assemblies, performance testing, and reassembly. The mean maintenance time (H_0) not including the time required to seal and unseal the indicator shall not exceed 240 minutes. The maximum maintenance time (H_1) shall not exceed 360 minutes not including the time required to seal and unseal the unit.

3.5.1.3 Reliability. The indicator shall have a minimum acceptable mean-time-between-failures (MTBF) of 1,000 hours at a 90 percent confidence level and a discrimination ratio of 1.5:1.

3.5.1.4 Service life. The indicator shall have a service life of 5,000 hours.

3.5.2 Construction. The indicator shall be so constructed that adjustments and repairs can easily be made by personnel of operating units and overhaul bases. The complete mechanism, consisting of all components except the pitch trim knob assembly and required lighting components, shall be removable from the case as a unit. The indicator shall be built to withstand the strains, jars, vibrations, and other conditions incident to shipment, storage, installation, and service.

3.5.2.1 Lubrication. Lubrication shall be in accordance with MIL-STD-838. The lubricant list shall be submitted for approval in accordance with MIL-STD-838 prior to qualification. Silicone lubricants shall be avoided where practical. When silicone lubricants must be used, justification detailing why they are required rather than other lubricants shall be provided with the submission of the lubricant list.

3.6 Performance. The indicator shall be capable of withstanding the following conditions:

a. Temperatures ranging from -54° to +71°C

b. Pressures ranging from 30 inches Hg down to 1.325 inches Hg (sea level to 70,000 feet), and external pressures up to 28 psia at 25°C

c. Relative humidity up to 95 percent

d. Fungus growth as encountered in tropical climates

e. Simulated rainfall as encountered in any locale

f. Exposure to simulated salt sea atmosphere for 50 hours

g. Sand and dust particles as encountered in desert areas

h. Acceleration forces up to 20g

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i. Vibration with a double amplitude of 0.1 inch from 5-14 Hz, ±1g from 14-23 Hz .036 inch double amplitude from 23 to 33 Hz and ±2g from 33 to 50 Hz

j. Voltages and frequencies varying from 103 to 127V and 380 to 420 Hz, respectively

k. Abrupt temperature change from +85° to +5°C

l. Shock of 15g in all three axes for 11 milliseconds with the equipment operating and 20g for 11 milliseconds when the equipment is not operating.

3.6.1 Radio noise interference. The indicator shall meet the radio noise interference requirements of MIL-I-6181.

3.6.2 Sensitivity. Each axis shall have a sensitivity of 0.25°.

3.6.3 Power consumption. The primary power consumption shall not exceed 18 va and for red or white lighting shall not exceed 10 va.

3.6.4 Magnetic property. When the indicator is held in various positions magnetically east or west of and not more than 12 inches from the center of a free magnet in a magnetic field having a horizontal field intensity of 0.18 ±0.01 oersted, the maximum deflection of the magnet shall not exceed 5°.

3.6.5 Insulation resistance. The indicator shall withstand a potential of 500V applied between isolated pins and between pins and the indicator for 10 seconds. The resistance shall be not less than 10 megohms.

3.6.6 Fogging. There shall be no fogging of the cover glass when the indicator is energized for a minimum of 1 hour in a 71°C ambient temperature and the indicator face then rubbed with an ice cube for 1 to 2 minutes.

3.7 Part numbering of interchangeable parts. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The item identification and part number requirements of MIL-STD-100 shall govern the manufacturer's part numbers and changes thereto.

3.8 Display. The display shall consist basically of a bank angle dial, bank indexes, a miniature airplane, vertical and horizontal pointers, a displacement r inter, rate-of-turn needle, slip indicator, two alarm flags, power off flag, . a sphere background which shall be free to rotate about the pitch, roll, and azimuth axes. Arrangement of the display shall be in accordance with figure 1 or 2, as applicable. All interior visible parts shall be finished in durable lusterless black color No. 37038 as approved by the qualifying activity.

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3.8.1 Bank angle dial. The bank angle dial dimensions and markings shall be in accordance with figure 1 or 2, as applicable, and figures 3 and 4. The distance between the front of the inner edge of the dial and the inner surface of the cover glass shall not exceed 3/4 inch. The portion of the dial within ±45° of the zero bank index shall be a portion of a cone with the visible portion of the dial at an angle that is approximately 75° to the roll axis of the indicator. The distance between the front edge of the outer periphery of the dial within ±45° of the zero bank index shall be not more than 11/16 inch from the inner surface of the cover glass. With respect to the remaining portion of the dial, the distance between the front edge of the outer periphery of the dial shall be not more than 1 inch from the inner surface of the cover glass. The outer radius of the dial shall be not less than 1-29/32 inches.

3.8.1.1 Dial masks. A mask shall be attached to each side of the bank scale dial in accordance with figures 4 and 5. The mask on the left shall also be the scale for the displacement pointer specified in 3.10.9.4. The surface of the scale shall follow the arc of the displacement pointer movement. The mean depth of the scale shall be approximately 1/4 inch from the cover glass and shall be symmetrical with the one on the right. The surface of the mask on the right shall be approximately 1/4 inch from the cover glass. This mask shall be the shield behind which the vertical pointer is driven as specified in 3.10.9.1. The inner vertical edges of the masks shall be not less than 1-29/32 inches from the center of the display. The mask shall be so designed that the displacement pointer can be driven out of view to the top as shown on figure 1.

3.8.1.2 Mechanism mask. A mechanism mask shall be attached to the roll gimbal and shall extend essentially from the bank dial to the sphere. The visible surface of the mask shall be shaped like the surface of a frustum of a cone. The large diameter shall be forwarded and shall be approximately 3-9/32 inches. The edge of the mask shall be hidden by the inner diameter of the bank scale. The small diameter shall be toward the rear and shall be approximately 3-5/32 inches. There shall be approximately 1/16 inch between this edge of the mask and the sphere.

3.8.2 Bank indexes. Bank indexes shall be affixed to and shall rotate with the roll gimbal. They shall be in accordance with figures 3 and 4. At zero roll indication, one shall be adjacent to and aligned with the zero roll marking on the bank dial. The other index shall be at the 180° roll position.

3.8.2.1 Bank angle command pointer (ARU-31/A only). The bank angle command pointer shall be in accordance with figure 7 The pointer shall be lusterless yellow, color No. 33793 (see 3.17). The bank angle command pointer shall be gear driven and adjustable ±62° from the zero roll indication. Adjustment shall be accomplished by means of an external knob in accordance with figures 2 and 10. The gearing shall have mechanical stops at ±62° +2° -0° and the drive knob

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shall contain a slip clutch to eliminate damage when the stops are contacted. The bank angle command pointer shall travel in a plane as close as is practical to the roll scale to minimize parallel error. It shall travel paralled to and follow the curvature of the bank angle scale. A torque of not less than 2 inch-ounces nor more than 12 inch-ounces shall be required to rotate the adjustment knob at any settable indication. A counterclockwise rotation of the knob shall result in a clockwise movement of the pointer and vice versa. Three rotations ±0.1 rotation of the knob shall be required to move the pointer from counter-clockwise stop to clockwise stop. The knob shall rotate with constant torque without evidence of binding or erratic movement over the complete range of settable indications. The knob shall be lusterless gray, color No. 36231. The knob shall have an appearance similar to the pitch trim knob, and its size shall be identical to the pitch trim knob.

3.8.3 Miniature airplane. The miniature airplane shall be in accordance with figure 6. The indicator shall be so designed that the miniature airplane will be located between the sphere and the horizontal pointer at the zero pitch position. The front surface of the center hairline shall be not more than 0.060 inch maximum from the sphere surface. The center of this center point shall be 2-1/2 ±1/32 inches from the top edge of the mounting flange. The wings of the miniature airplane shall be curved so that they follow the swing radius of the vertical pointer.

3.8.4 Vertical and horizontal pointers and alarm flags. The vertical and horizontal pointers shall provide flight director information as specified herein and shall be in accordance with figures 1 and 6. Alarm flags shall be provided for the displacement, horizontal, and vertical pointers. A typical flag is illustrated on figures 1 and 7.

3.8.5 Slip indicator and rate-of-turn display. A slip indicator and rate-of-turn display shall be provided in accordance with figures 1 and 8.

3.8.6 Gimbal system. The gimbal system shall consist of three gimbals having axes that are mutually perpendicular.

3.8.6.1 Roll gimbal. The roll gimbal shall be the outer gimbal. The axis of the gimbal shall be perpendicular to the face of the indicator and in line with the center of the sphere. The gimbal shall be driven by the roll servo mechanism and shall be designed to mount the bank indexes and pitch gimbal.

3.8.6.2 Pitch gimbal. The pitch gimbal shall be suspended in the roll gimbal. The axis of the gimbal shall be at right angles to the roll gimbal axis and in line with the center of the sphere. The gimbal shall be driven by the pitch servo mechanism. The azimuth mechanism shall be mounted on the pitch gimbal.

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3.8.6.3 Azimuth gimbal. The azimuth gimbal mechanism shall consist of a shaft and the azimuth servo mechanism. The shaft shall be so mounted as to be perpendicular to the pitch gimbal axis and, at zero pitch attitude, perpendicular to the roll gimbal axis. The shaft shall support the two sphere halves at their centers. The shaft shall be driven by the azimuth servo mechanism.

3.8.6.4 Indicator spheroid. The indicator spheroid shall contain the azimuth servo for driving two sphere caps for heading information. The indicator shall be so mechanized that when a heading signal is received for a right-to-left change in azimuth, the indicator spheroid will rotate left to right with respect to the indicator bezel when viewed from the front of the cover glass. The sphere shall have a fixed segment, with respect to the azimuth axis, around the outer surface of the sphere. This fixed surface shall define the horizontal reference of the indicator spheroid. The azimuth servo loop shall drive both hemispheres which form the sphere and which intersect the fixed surface at the horizontal reference line. The spheroid markings shall be in accordance with figure 9.

3.8.6.5 Segmented sphere. The sphere shall form the background of the display. It shall be divided into three sections marked as shown on figure 9. Essentially, these sections will be the top and bottom halves of the sphere and a narrow center band. This band is for masking the clearance between the top and bottom sections of the sphere. The clearance is required to bring out the mechanical supports and electrical leads from inside the sphere. The sphere as a whole shall rotate about the roll and pitch axes with continuous freedom in either direction. It shall rotate with the roll gimbal about the roll axis for roll indications and shall rotate in relation to the roll gimbal about the pitch axis for pitch indications. The roll axis shall be perpendicular to the face of the indicator and the pitch axis. The pitch axis shall intersect the roll axis in the center of the sphere and shall rotate with the roll gimbal. The top and bottom sections of the sphere are the azimuth caps. These shall rotate with respect to the center band about the azimuth axis as a unit. The azimuth axis shall be perpendicular to the pitch and roll axes. The distance between the cover glass and the sphere shall be held to a minimum and shall not exceed 7/16 inch. The distance between the segmented sphere and each of the sphere caps shall be 0.025 ±0.010 inch.

3.8.7 Outline dimensions. The indicator, with amplifier and associated display, shall be in accordance with figure 10. The location of the elapsed time indicator shall be subject to the approval of the qualifying activity prior to fabrication and production.

* 3.9 Pitch trim. A pitch trim knob shall be incorporated for adjusting the position of the sphere and shall be mounted on the lower right side of the face of the indicator and marked in accordance with figures 1 and 10. The knob shall have approximately 1/2 turn of freedom in the clockwise (cw)

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direction and 1/4 turn of freedom in the counterclockwise (ccw) direction from the zero pitch trim position. Freedom in each direction shall be limited by a positive stop. When the knob is rotated to the stop position in the cw direction, the sphere shall rotate to deflect the horizon line upward to indicate a dive between 10° and 20°. When the knob is rotated to the stop in the ccw direction, the sphere shall rotate to deflect the horizon line downward to indicate a climb between 5° and 10°. The servo system shall operate properly for any pitch trim adjustment. A torque of not less than 1 inch-ounce nor more than 8 inch-ounces shall be required to adjust the pitch trim knob. The knob shall move smoothly without evidence of binding or erratic movement. The internal design of the indicator shall be such that any pitch trim input will fade out automatically as the sphere is driven to the extreme pitch indication; the sensing of the trim signal will be reversed to correspond to the side of the sphere showing when the sphere is driven through the 90° pitch position. Fadeout shall occur as the sphere is driven from the 20° to 80° pitch indications and fade-in shall occur as the indicated pitch angle changes from 80° to 20°. It shall occur on both sides of the sphere for both climb and dive indications. The design shall be such that no degradation of performance will occur at any pitch or roll trim setting. The rotation of the pitch trim knob from full cw to full ccw should produce a smooth, continuous variation of pitch proportional to the knob rotation.

3.9.1 The design shall be such that when two indicators are connected in parallel, a change in pitch trim adjustment of one indicator will not affect the indication on the second instrument. The pitch trim knob, item H, figure 1, shall be lusterless gray, color No. 36231. The arrow provided on the knob shall be lusterless black, color No. 37038, and the reference mark against which the arrow on the knob is set shall be lusterless white, color No. 37875.

3.10 Indicator synchro control transformer. The synchro control devices on the indicator axes shall be high-impedance Eclipse-Pioneer AY500-5 synchros, or equal. The servo circuit shall be so designed that when one to four indicators per axis are connected to a single test transmitter synchro, no apparent change will be noted in performance. All synchro control device stator leads shall be electrically isolated from each other, from other circuitry in the case, and from the case.

3.10.1 Standard test transmitter synchro. The standard test transmitter synchro shall be an Eclipse-Pioneer AY201-1, or equal, with minimum and known errors. These errors shall be subtracted from those of the indicator under test.

3.10.1.1 The transmitter shall be equipped with an accurate dial having a ±1° vernier and readable to 0.1°. It shall be graduated through 360° and shall be capable of continuous rotation.

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3.10.1.2 The transmitter rotor and stator leads shall be identified and calibrated as:

a. Rotor leads designated H and C, with C grounded

b. One stator lead designated Z and connected to C

c. 26V, 400 Hz applied across H and C and the voltage measured across the two remaining stator leads and across H and one of the stator leads

d. The rotor turned until the voltage across H and the one stator lead is approximately maximum (about 35V) and the voltage across the two stator leads is exactly minimum. The dial locked to the rotor with the dial exactly on zero.

e. The dial turned through increasing numbers and the voltage measured across the two undesignated stator leads and Z. The voltage across XZ shall increase before decreasing and the voltage across YZ shall decrease before increasing.

3.10.1.3 Test transmitter excitation. The test transmitter rotors shall be excited from the same power-supply voltage as that operating the indicator. It shall be so applied that the voltage from rotor lead C to rotor lead H will be in phase with the voltage from pin A to pin B of the indicator with lead C and pin A (see table I) common.

3.10.2 Indication zeroing and sensing. When the pitch and roll servo control devices of the indicator are connected to standard test transmitters in accordance with table I, with the respective transmitters set at the 180° dial reading, The indications shall come to rest at zero. Increasing dial readings of the respective test transmitter shall produce right-roll and pitch-up indications. Decreasing dial readings shall produce opposite indications. When the azimuth servo control transformer of the indicator is connected to a standard test transmitter in accordance with table I, with the transmitter set at the 0° dial reading, the indicator azimuth caps shall come to rest at the 0° azimuth indication. Increasing dial readings of the test transmitter shall result in the visible portion of the azimuth caps moving from right to left giving indication of right turns. Decreasing dial readings of the test transmitter shall result in a left-turn indication.

3.10.3 Attitude actuation. The sphere shall be rotated about the pitch, roll, and azimuth gimbal axes by means of geared servo motors until a null is attained between the control transformer synchros on the indicator axes and the transmitter synchros in the pitch, roll, and azimuth reference. The freedom of sphere rotation about any axis shall not be limited.

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TABLE I. Indicator Connections

(Connector shall mate with Bendix PT06A-18-32S electrical receptacle, or equal)

Connector pins	Function
A	Neutral (external power ground)
B	115V, 400 Hz
C (+)	Horizontal pointer alarm flag
D (−)	
E	Internal lighting power input for white lighting (0-5V)
j	Power warning flag Connect to pin A to complete circuit
g	Neutral for internal lighting
h	Internal lighting power input for red lighting (0-5V)
d (X)	Roll servo control transformer stator leads
e (Y)	
f (Z)	
a (X)	Pitch servo control transformer stator leads
b (Y)	
c (Z)	
Y (+)	Vertical pointer (deflects to right)
Z (−)	
W (+)	Horizontal pointer (deflects downward)
X (−)	
U (+)	Vertical pointer warning flag (out-of-view)
V (−)	
S (+)	Displacement pointer (deflects downward)
T (−)	
P (+)	Rate-of-turn indicator (deflects to right)
R (−)	
L	Spares
M	
N	
J (+)	Displacement pointer flag (out-of-view)
K (−)	
F (X)	Azimuth servo control transformer stator leads
G (Y)	
H (Z)	

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3.10.4 Followup rate. The maximum rate at which the motors drive the gimbals shall be not less than 300° per second in roll or azimuth nor less than 90° per second in pitch.

3.10.5 Accuracy. The pitch and roll indications shall agree with the input signals from their respective test transmitter synchros to within ±0.5° at indications within ±30° and to within ±1° at all other indications. The azimuth indication shall agree with the input signal from its test transmitter synchro to within ±1° at all indications.

3.10.6 Followup operation. The electrical and mechanical design of the indicator shall be such that in the process of followup, indications of pitch, roll, and azimuth will move smoothly without hunting, oscillating, or jumping more than 0.125° total amplitude when each servo input is connected to standard transmitters for testing, or when each servo input axis receives a standard synchro input which simulates the motion of an aircraft. The input rates shall be between 0° per second and the maximum required for the respective axis. No interaction shall occur when each axis is driven independently or when two or three axes are driven simultaneously from 0° per second to their respective maximum rate. There shall be no sustained electrical or mechanical oscillations in any axis at any attitude. The indicator shall stabilize at the ordered position without further oscillations or overshooting within 0.20 seconds after application of a step input equal to the maximum allowable followup lag. There shall be no ambiguity in pitch, roll and azimuth when the transmitter signal is instantaneously switched from 0° to 180°.

3.10.6.1 Amplifiers. The amplifiers provided as a part of the indicator shall be three-stage, transistorized amplifiers with redundant transistor stages. The external configuration of the amplifier shall conform to figure 11. The amplifier shall incorporate redundant circuits so that the indicator will continue to operate, though with reduced performance, should partial failure of the applicable amplifier occur. The amplifier shall be housed in a separate package from the indicator proper so that it can be physically attached to the rear of the indicator or mounted remotely. It shall be so designed that it will meet the requirements of this specification when mounted directly to an airframe without shock mounts as well as when mounted on the indicator. Each amplifier channel shall be a separate module within the amplifier case; these channels shall be interchangeable. The amplifier unit need not be hermetically sealed but shall be sealed in such a manner as to withstand all environmental tests specified herein.

3.10.6.1.1 Zeroing adjustments. Means shall be incorporated to externally adjust the position of the sphere ±3° minimum in pitch, roll, and azimuth. The initial setting of the indications at the time of delivery of the indicator shall indicate zero within ±0.5° when each axis of the indicator is connected to a test transmitter. The pitch adjustment shall be combined with the

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pitch-trim circuit by means of a settable knob. The roll and azimuth adjustments shall be on the rear of the amplifier case, shall be clearly marked, and shall be mounted to the indicator. The trim adjustments shall be equipped with stops so positioned that erroneous indications of attitude cannot result from positioning of these adjustments. The trim adjustments shall incorporate locknuts or similar devices to secure the adjustment setting.

3.10.6.1.2 Power off flag. A power off flag shall be incorporated on the lower left (ARU-21/A or ARU-11/A) or lower right (ARU-31/A) of the face of the indicator as shown on figures 1, 2, and 7. The word OFF shall be lusterless black, color No. 37038, on a lusterless red background conforming to color No. 38905. A lead shall be brought out through the electrical connector as specified in table I. When either the ground connection or power is removed, or both, the warning shall appear. When power is applied but internal failure of the B+ amplifier voltage occurs, the warning shall appear.

3.10.6.1.3 Amplifier pin connections. The connections between the amplifier and the indicator proper shall be made in accordance with table II. The manufacturer shall supply the qualifying activity with a table of functions of all pin connections made between the indicator and amplifier. This data shall be supplied at a time designated by the qualifying activity. The position of the polarizing key for the amplifier connector shall be subject to approval by the qualifying activity.

TABLE II. Amplifier Connector

(Connector shall mate with Bendix clamp PT01E-16-26P (SR), or equal, or Potting PT1P-16-26P, or equal)

Connector pins	Functions
A	115V a-c amplifier power supply
B, C, E, J, K, L	Reserved for signal level voltages
M	Ground
T, Z, a, P	Reserved for signal level voltages
Note Pins not specified above may be used as required	

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3.10.7 Electrical power. Power supplied to the indicator shall be 115V, single-phase, 400-Hz in accordance with MIL-STD-704.

3.10.7.1 Electrical power variation. The indicator shall operate properly for any ac voltage between 103 and 127V, for any frequency between 380 and 420 Hz or for any combination thereof.

3.10.7.2 Case isolation. The case shall be isolated from all electrical circuits therein.

3.10.8 Slip indicator. The slip indicator shall be incorporated into the display in accordance with figures 1 and 8.

3.10.8.1 Tube. The glass of the slip indicator tube shall be made of clear annealed glass tubing free from flaws that will seriously affect readability of the slip indicator. The inside of the tube shall be smooth and uniform so that the ball will roll freely except for damping of the liquid. The arc length of the visible portion of the tube shall be approximately 1-1/4 inches. The radius of curvature of the tube shall be 4-1/2 inches.

3.10.8.2 Ball. The ball shall be a highly polished, accurate, 13/64-inch white sphere. A minimum of two-thirds of the ball shall remain in view at either extreme of the tube when viewed from a position 12 inches in front of the zero mark of the slip indicator at an angle of 30° with the horizontal plane prescribed by the base of the indicator.

3.10.8.2.1 Sensitivity. The ball shall roll freely to the end of the tube when the indicator is rotated to a bank angle of ±7° ±2°. The ball shall not stick during rotation or at either end of the tube.

3.10.8.2.2 Zero position. The ball shall indicate zero within ±1/32 inch when the indicator is mounted in its normal level position.

3.10.8.2.3 Friction. The ball shall roll smoothly without sticking when the indicator is rotated in its vertical plane at a rate of 2.0° +0.1° -0.0° per second. The ball shall return to zero within ±1/32 inch when the indicator is returned to a level position from an angle of ±10° at the above rate.

3.10.8.2.4 Damping. The time required for the ball to roll from zero to the end of the tube shall be between 0.15 and 0.50 second when measured in accordance with 4.6.35.

3.10.8.3 Damping liquid. The damping liquid used shall be sufficiently colorless to preclude interference with visibility of the ball under daylight and night lighting conditions and under normal temperature and temperature extremes.

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3.10.8.4 Slip indicator filling. With all air trapped in the expansion chamber and the indicator in a level position, no part of the air bubble shall be visible as the indicator is rotated to an angle of 12° with the expansion chamber low and viewed as specified in 3.10.8.2.

3.10.9 Flight director display. The flight director display shall consist of a vertical pointer, a horizontal pointer, a pointer to show displacement from commanded flight path, and two signal level alarm flags. The pointers and flags shall be driven by meter movements and shall be in accordance with figures 1, 5, 7, and 8. The design of the pointer meter movements shall be such that when not energized, the pointers will be maintained, as near as practicable, to their adjusted zero positions while under all flight conditions, including rough air. All meter movement leads shall be electrically isolated from each other, from other circuitry in the case, and from the case. All meter movements shall operate as specified herein when operated singly, when connected in parallel with a like meter movement, or when connected in parallel with a 1,000-ohm load to a signal source that has an impedance between 350 and 1,000 ohms. The yellow area of the horizontal and vertical steering pointers shall be centered over the tail of the miniature airplane.

3.10.9.1 Vertical pointer. The vertical pointer shall be positioned directly behind the indicator light wedge in front of the miniature airplane. It shall be mounted on a meter movement that is pivoted at a point approximately 4-1/2 inches behind the pointer in such a manner that the pointer will remain vertical at zero center or at any point of deflection to the right or left of center, and will deflect right or left in agreement with the polarity of direct current received from the flight director computer. An external mechanical zero adjustment shall be provided. The portion of the vertical pointer dimensioned in figure 7 shall be lusterless yellow, color No. 33695.

3.10.9.1.1 Deflection. The design of the vertical pointer shall be such that the deflection of the pointer to the left of the center position will be limited to 1 inch by a physical stop and the maximum deflection to the right will carry the pointer behind the mask on the right side of the display. The direction of deflection for a given input signal shall be as specified in table I.

3.10.9.1.2 Response. The vertical pointer response shall be linear in degree with respect to current within 7.5 percent of the proportionate full-scale value. A deflection of 7/8 inch, measured from the center of the dial to the pointer along the horizontal line, shall require 2.2 ma ±7.5 percent; 2.5 ma shall not deflect the pointer by more than 1-1/16 inches. An application of 10 ma of the proper polarity (as specified in table I) shall drive the pointer from view to behind the shield at the right. The application of 10 to 17 ma as a step input to drive the pointer from view shall not damage the meter movement.

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3.10.9.1.3 Resistance. The resistance across the vertical pointer input terminals shall be 1,000 ohms ±3 percent at 25°C.

3.10.9.1.4 Damping. The pointer mechanism shall be so damped that there will be no more than 1-1/2 percent overshoot.

3.10.9.1.5 Response time. Response time of the vertical pointer shall be a maximum of 1/3 second. Response time is the time required for the pointer to reach 90 percent of its final indication as counted from the instant of application of the current. The pointer shall be considered as having come to apparent rest when it has reached within ±1 percent of the scale length of the actual rest point.

3.10.9.2 Vertical and horizontal pointer alarm flags. The alarm flags shall be operated by a suppressed zero-type mechanism which shall hold the flag against a stop in the position shown on figure 1 in the absence of current or when the current application is below a predetermined value as specified in 3.10.9.2.1, 3.10.9.2.2, and 3.10.9.2.3.

3.10.9.2.1 Deflection. The alarm flags shall be designed to deflect out-of-view with the input signal polarity of table I connected to a voltage that is polarized plus and minus, respectively.

3.10.9.2.2 Response. The alarm flags shall leave their visible position stop with a current application of 180 microamperes (μa) of more. The flags shall disappear with a current application of 245 μa. Response of the alarm flag mechanisms shall be approximately linear in degree with respect to current. The application of 380 μa as a step input shall not damage the meter movement.

3.10.9.2.3 Resistance. The resistance across the alarm-flag inputs shall be 1,000 ohms ±3 percent at 25°C.

3.10.9.3 Horizontal pointer. The horizontal pointer shall be positioned directly behind and as near as practicable to the vertical pointer. The horizontal pointer shall be mounted on a meter movement which shall be pivoted at a point approximately 4-1/2 inches behind the pointer. With the indicator in normal operating position, the pointer shall move in such a manner that it will remain horizontal at zero center or at any point of deflection up or down from center, and shall deflect up or down in agreement with the polarity of direct current received from the flight director computer. The dimensioned portion of the horizontal pointer shown on figure 7 shall be lusterless yellow, color No. 33695. An external mechanical zero adjustment shall be provided.

Note: The flight director computer referred to herein is designed so that current to the pointer movements is limited to 2.5 ma maximum when in a mode where the pointers are being used. It will provide 10 to 17 ma of the proper polarity to either or both pointer movements when the pointers are to disappear.

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3.10.9.3.1 Deflection. The maximum deflection upward shall be physically limited to 1-1/16 inches. The design shall be such that the pointer can be deflected down (1-15/16 inches minimum) until it disappears from view behind a mask. This mask shall not be nearer than 1-29/32 inches to the roll axis of the indicator. The direction of deflection for a given input signal shall be as specified in table I.

3.10.9.3.2 Response. The horizontal pointer response shall be linear in degree with respect to current within 7.5 percent of the proportionate full-scale value. A deflection of 7/8 inch, measured from the center of the dial to the pointer along a vertical line, shall require 2.2 ma ±7.5 percent, 2.5 ma shall not deflect the pointer by more than 1-1/16 inch. Application of 10 ma of the proper polarity shall drive the pointer from view behind its mask at the bottom of the display. The application of 10 to 17 ma as a step input to drive the pointer from view shall not damage the meter movement.

3.10.9.3.3 Resistance. Resistance across the horizontal pointer input terminals shall be 1,000 ohms ±3 percent at 25°C.

3.10.9.3.4 Damping. The pointer mechanism shall be so damped that there will not be more than 1.5 percent overshoot.

3.10.9.3.5 Response time. Response time (see 3.10.9.1.5) of the horizontal pointer shall be a maximum of 1/3 second.

3.10.9.4 Displacement pointer. The displacement pointer and scale shall be in accordance with figure 5. The pointer shall be mounted on a meter movement that is pivoted at a point approximately 4-1/2 inches behind the pointer. With the indicator in normal operating position, the pointer shall move in such a manner that it will remain horizontal at zero center or at any point of deflection up or down from center, and shall deflect up or down in agreement with the polarity of direct current received from the flight director computer. The pointer shall run adjacent to the displacement scale as shown on figure I. The design of the displacement pointer shall be such that application of 500 µa of the proper polarity applied as a step input will drive the displacement pointer out of view to the position shown on figure 1. Application of 1,000 µa to drive the pointer out of view shall not damage the meter movement. This pointer shall perform as specified herein when connected to a 350-ohm signal source impedance and when connected in parallel with two additional similar meter movements or two 1,000-ohm dummy loads. An external mechanical zero adjustment shall be provided.

3.10.9 4.1 Deflection. Direction of deflection for a given input shall be as specified in table I.

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3.10 9.4.2 Response. Displacement pointer response shall be linear in degree with respect to current within 7.5 percent of the proportionate full-scale value Deflection of 7/8 inch, measured from the zero position of the scale to the centerline of the pointer, shall require 150 µa ±7.5 percent.

3.10 9.4.3 Resistance Resistance across the displacement pointer input terminals shall be 1,000 ohms, ±3 percent at 25°C.

3.10.9 4.4 Damping. The pointer mechanism shall be overdamped (no overswing)

3.10.9.4.5 Response time. Response time (see 3.10.9 1.5) of the indicating pointer shall be a minimum of 1.15 seconds and a maximum of 2 seconds.

3.10 9 5 Displacement pointer flag. The displacement pointer flag shall be in accordance with figures 1, 5, and 7. Its characteristics shall be the same as those specified for the vertical pointer alarm flag (see 3.10.9.2) The displacement pointer flag terminals and polarization shall be as specified in table I.

3.10.10 Rate-of-turn. The rate-of-turn display shall be in accordance with figures 1 and 8. The moving index shall be mounted as a meter movement that is pivoted at a point approximately 4-1/2 inches behind the pointer. The design shall be such that the pointer is as stable as practicable under rough air conditions. In the normal operating position, the index shall move left or right in agreement with the polarity of the direct current received from the remote rate-of-turn sensor. The scale shall conform to the path of the index so that there is a minimum of parallax. The front surface of the scale and index shall be within 1/4 inch of the cover glass at zero displacement. The index deflection to either side of the center zero position shall be limited to 5/8 inch. At each of the fully deflected positions, a minimum of one-half of the index shall be in view when viewed at a horizontal distance of 2 feet from the center point of the display. An external mechanical zero adjustment shall be provided. Fixed mechanical stops shall be provided at the extreme deflection points to prevent pointer hangup on the case or mechanisms A stop or bumper shall be provided to prevent the pointer from moving up over the scale during shock or acceleration. The stop shall be effective at all pointer indications.

3.10.10.1 Deflection (index). The direction of deflection for a given input shall be as specified in table I.

3.10.10 2 Response (index). The index response shall be linear in degree with respect to current within 7.5 percent of the proportionate full-scale value. A deflection of one index width from the zero position shall require 0.5 ma ±7.5 percent. A deflection of two index widths so that the index aligns with the appropriate outer scale mark shall require 1 0 ma ±7.5 percent

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3.10.10.3 Resistance (index). Resistance across the rate-of-turn input terminals shall be 1,000 ohms ±3 percent at 25°C.

3.10.10.4 Damping (index). The pointer mechanism shall be so damped that there will be no more than 1.5 percent overshoot.

3.10.10.5 Response time (index). Response time (see 3.10.9.1.5) of the rate-of-turn index shall be a maximum of 1/3 second.

3.10.11 Elapsed time indicator. An elapsed time indicator in accordance with MS17322 shall be incorporated to record the running time of the indicator proper. The elapsed time indication shall be provided in such a manner that it will be unnecessary to unseal the indicator proper to read the elapsed time recorded. The location shall be subject to the approval of the qualifying activity.

3.11 Electrical connectors. A Scintilla Division, Bendix Aviation Corporation Part No. 10-150631-32P connector, or equal, shall be provided on the indicator in the manner shown on figure 10. Connections shall be in accordance with table This connector shall mate with a Scintilla Division Part No. 10-150606-32S connector, or equal. A second connector, as shown on figures 10 and 11, shall be pr vided on the indicator to interconnect the indicator proper with the amplifier. The connector on the indicator shall be a Scintilla Division Part No. 10-150630-26P connector, or equal, which shall mate with a Scintilla Division Part No. 10-179145-26S connector, or equal, to be supplied as part of the amplifier. The physical position of each electrical connector and location of the polarizing key on the connector with respect to the indicator case shall be as shown on figure 10.

3.12 Cover glass and lighting wedge. The cover glass or lighting wedge shall be flat and free from discolorations, scratches, and striae that might interfere with reading of the indicator. The glass shall be tempered, shall be mounted in the flange by insertion within the case, and shall be a minimum of 3/16 inch thick. All reflecting glass surfaces shall be provided with a reflection-reducing coating that meets the requirements of MIL-C-14806 in addition to withstanding the environmental conditions specified herein.

3.13 Indicator case. The indicator case shall be in accordance with MS33545 with the exceptions stated herein. Dimension A of MS33545 shall be in accordance with figures 1 and 10. Dimension L shall be the length of the envelope when the amplifier is attached physically to the indicator proper and shall not exceed the 8 inches required for electrical connectors as shown on figure 10. The pitch trim knob shall be approximately 3/4 inch in diameter and 1 inch long. The indicator case shall be hermetically sealed.

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3.13.1 Filling medium. The filling medium shall be at least 98 percent pure, free of dust particles, and shall contain not more than 0.006 milligram of water vapor per liter (dewpoint -65°C) at the filling pressure. The filling medium shall be a mixture of 88 to 92 percent nitrogen and the remainder helium or 95 percent helium and 5 percent carbon dioxide. The absolute pressure of the filling medium in the case shall be 12 to 14.7 psi.

3.13.2 Leak rate. The indicator case shall be so sealed that the maximum leak rate will not permit more than 10 percent contamination of filling medium after 1,000 hours at a pressure differential of approximately 12 to 14.7 psi.

3.14 Lighting. Internal lighting shall be designed into the indicator. The design of the indicator shall be such that the internal lighting may be either white or red depending upon the pins (specified in table I) to which the power for lighting is applied. With the exceptions specified herein, the red lighting shall be in accordance with MIL-L-25467 and the white lighting shall be in accordance with MIL-L-27160, except that the miniature airplane brightness shall range between 1.0 and 2.0 foot-lamberts. The ratio of the brightness of the gray portion of the sphere to the brightness of the miniature airplane shall never exceed 0.6. The internal lighting of the prime standard and all qualification indicators shall be subject to the approval of the qualifying activity.

3.14.1 Lamps. Lamps shall be in accordance with MIL-L-25467 and MIL-L-27160.

3.14.2 Prime standard. The internal lighting shall be comparable to and shall balance in brightness with a prime standard indicator. The prime standard shall be either an attitude director conforming to the requirements of this specification or a lighted model. The prime standard shall be equipped with both red and white lighting and shall be used as the reference for acceptance of all production indicators.

3.14.2.1 Prime standard recertification. All representative areas of the face of the prime standard indicator shall be measured for brightness in foot-lamberts. Each area shall be measured at a sufficient number of locations to obtain data representative of each area including the brightest and dimmest portions of the indicator in each display color. The prime standard indicator shall be measured periodically (at least every 50 hours of operation or annually, whichever occurs first) to determine any brightness change. The data and, if required by the qualifying activity, the prime standard shall be submitted for evaluation and reapproval or corrective action, or both.

3.14 2.2 Visual comparison. The lighting level of the indicator shall fall within the limits of the prime standard when energized at the voltage levels specified in 4.6.22.2.2.

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3.15 Bearings

3.15.1 Motor generator and synchro. Motor generator and synchro bearings shall be in accordance with MIL-B-83292.

3.15.2 Jewel. Synthetic jewel bearings shall be in accordance with MIL-B-27497.

3.16 Weight. The weight of the indicator with amplifier attached shall not exceed 9 pounds.

3.17 Colors. All colors specified herein shall be in accordance with FED-STD-595.

3.18 Finishes and protective coatings

3.18.1 Aluminum alloy parts. Aluminum alloy parts shall be covered with an anodic film conforming to MIL-A-8625, except:

a. Dials, small holes, and case inserts need not be anodized

b. Aluminum alloys which do not anodize satisfactorily shall be coated with a chemical flim in accordance with MIL-C-5541

c. Where the primary purpose of the treatment is to afford a suitable paint base, chemical treatments in accordance with MIL-C-5541 may be used in lieu of anodizing

d. Castings containing nonaluminum integral inserts may be treated with a chemical flim in accordance with MIL-C-5541 in lieu of anodizing

e. When abration resistance is a factor, chemical films in accordance with MIL-C-5541 shall not be used in lieu of anodizing

f. Parts in hermetically sealed containers need not be anodized

g. When the part is plated with tin over a copper flash, the part need not be anodized.

3.18.2 Cadmium plating. Cadmium plating shall be in accordance with QQ-P-416 type II or III, as applicable, and of a class that is adequate to achieve the degree of protection required.

3.19 Markings. Unless otherwise specified, the dimensions of all markings on the indicator shall be in accordance with figures 1 through 9. All visible portions of the indicator shall be lusterless black, color No. 37038

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3.19.1 Numerals and letters. Unless otherwise specified, all numerals and letters on the indicator, other than the nameplate, shall be in accordance with MS33558.

3.19.2 Readability. All markings shall be clear and sharp. Dimensions shall be as specified and on any given mark the width shall not vary noticeably throughout the length of the mark All markings which for some given indication become adjacent and are specified to have the same basic dimension at the adjacent point shall not be noticeably different at that point.

3.20 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

3.21 Workmanship. The indicator, including all parts and accessories, shall be fabricated and finished in a thoroughly workmanlike manner. Particular attention shall be given to freedom from blemishes, defects, burrs, and marking of parts and assemblies, thoroughness of soldering, welding, brazing, painting, wiring, riveting, alignment of parts, and tightness of assembly screws and bolts

3.21 1 Riveting. Riveting operations shall be carefully performed to ensure that the rivets are tight and satisfactorily headed.

3.21 2 Cleaning. The indicator shall be thoroughly cleaned, and loose, spattered, or excess solder, metal chips, and other foreign material removed during and after final assembly.

3 21 3 Dimensions and tolerances. Dimensions and tolerances not specified shall be as close as is consistent with the best shop practices. Where dimensions and tolerances may affect the interchangeability, operation, or performance of the indicator, they shall be held or limited accordingly.

3.21 4 Screw assemblies. Assembly screws and bolts shall be tight. The word tight means that the screw or bolt cannot be appreciably tightened further without damage to the screw or bolt threads.

4. QUALITY ASSURANCE PROVISIONS

4 1 Responsibility for inspection Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 Classification of tests. The inspection and testing of indicators shall be classified as:

a. Qualification testing . . . . . See 4.4

b. Quality conformance tests . . . . . See 4.5.

4.3 Test conditions

4.3.1 Standard atmospheric conditions. Unless otherwise specified, tests shall be conducted at atmospheric pressure (approximately 29.92 inches Hg) and at room temperature (approximately 25°C).

4.3.2 Voltage input. Unless otherwise specified, the indicator shall be tested with 115V, 400-Hz (nominal), single-phase power.

4.3.3 Proper connection. When the indicator is referred to as being properly connected, it shall mean that it is connected to a test fixture that provides a lighting circuit voltage variable from 5V ac or dc to approximately zero volts, with synchro signals in accordance with 3.10 through 3.10.2, variable current dc signals, a 1,000-ohm source, with reversible polarity to the meter movements, ac power as required, and that pin J is jumped to pin A through a closed switch.

4.3.4 Visual observations. Unless otherwise specified, all visual observations of indicator performance shall be made at a distance of approximately 2 feet.

4.3.5 Indicator. Unless otherwise specified in a particular test, the word indicator as used herein shall mean the indicator proper together with the amplifier unit.

4.3.6 Recordings. Upon application of power, the indication of the elapsed time indicator shall be recorded. Readings shall be recorded after completion of all tests specified under 4.6.

4.4 Qualification testing

4.4.1 Test samples. The test samples shall consist of nine indicators representative of the production equipment. Three of the samples shall be subjected to all the tests except 4.6.53 and 4.6.54 and the other six samples shall be subjected only to the test specified in 4.6.53. The samples shall be identified with the manufacturer's part number and such other information as required by the qualifying activity.

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4.4.1.1 Data to accompany test samples. Unless otherwise specified, the test samples shall be accompanied by:

a. Brief operating data to enable test personnel to correctly operate the equipment

b. Engineering data in the form of assembly or detail drawings (2 sets).

4.4.2 Test report, test samples, and data for the qualifying activity. When the tests are conducted at a location other than the laboratory of the qualifying activity, the following shall be furnished to that activity.

a. Test report - Three copies of a test report in accordance with MIL-STD-831

b. Test samples - The test samples that were tested and an untested sample

c. The data required under 4.4.1.1.

4.4.3 Qualification tests. The qualification tests shall consist of all the tests specified under 4.6, except 4.6.54.

4 5 Quality conformance tests. The quality conformance tests shall consist of:

a. Individual tests. . . . . See 4.5.1

b. Sampling tests. . . . . See 4 5.2.

4.5.1 Individual tests. Each indicator shall be subjected to the following tests as described under 4.6:

a. Examination of product . . . . . See 4.6.1

b. Early failure detection test . . . . . See 4.6.2

c. Starting . . . . . See 4.6.3

d. Pitch and bank zero. . . . . See 4.6.4

e. Sensitivity. . . . . See 4.6.5

f. Followup rate test . . . . . See 4.6.6

g. Followup accuracy. . . . . See 4.6.7

h. Followup operation . . . . . See 4.6.8 through 4.6.8.4

i. Gimbal freedom . . . . . See 4.6 9

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j. Pitch trim . . . . . See 4.6.10

k. Pitch trim fade. . . . . See 4.6.10.2

l. Pitch trim zero. . . . . See 4.6.10 3

m. Zeroing adjustments. . . . . See 4.6.11

n. Sealing. . . . . See 4.6.12

o. Power off flag . . . . . See 4.6.13

p. Slip indicator sensitivity . . . . . See 4 6 14

q. Slip indicator zero position . . . . . See 4.6.15

r. Slip indicator friction. . . . See 4 6 16

s. Slip indicator visibility. . See 4.6 17

t. Slip indicator filling . . . See 4.6.18

u. Response . . . . . See 4 6 19

v. Pointer and flag balance . . . See 4 6 20 and 4 6 20.1

w. Pointer alarm flags. . . . . See 4.6 21

x. Lighting . . . . . See 4.6 22

y. Fogging . . . . . See 4.6 23

z. Bank angle command pointer adjustment (ARU/31 only) See 4.6.24 and 4.6 24.1.

4.5.2 Sampling tests

4.5.2 1 Sampling plan A One indicator shall be selected at random from each 100 or less produced on the contract or order and subjected to the following tests as described under 4 6

a. Individual tests . . . . . See 4 5 1

b. Power consumption . . . See 4 6 25

c. Low voltage . . . . . See 4.6.26

d. Magnetic property . . . . . See 4.6.27

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e. Acceleration (operating) . . . . . See 4.6.28

f Insulation resistance . . . . . See 4.6.29

g. Vibration error . . . . . See 4.6.30

h. Horizon line and bank index null position shift . . . . . See 4.6.31

i. Low temperature operation . . . . . See 4.6.32

j. High temperature operation . . . . . See 4 6.33

k Slip indicator leakage . . . . . See 4.6.34

l. Slip indicator damping, room temperature . . . . . See 4.6.35

m. Response time. . . . . See 4.6.19.1.

4.5 2.2 Sampling plan B. Unless otherwise specified (see 6.2), 3 indicators shall be selected at random from the first 15 items on the contract or order and subjected to the following tests in the order specified and as described under 4.6

a Sampling plan A tests . . . . . See 4 5.2.1

b Lighting circuit test . . . . . See 4.6.36

c. Radio noise interference . . . . . See 4.6.37

d High temperature exposure . . . . . See 4 6 38

e. Temperature shock . . . . . See 4.6.39

f Temperature-altitude (cycling) . . . . . See 4.6.40

g Sealing thermal shock . . . . . See 4 6 41

h Dust . . . . . See 4 6.42

i Humidity . . . . . See 4 6.43

j Fungus . . . . . See 4 6.44

k Salt fog . . . . . See 4.6.45

l. Rain . . . . . See 4.6.46

m. Acceleration (nonoperating) . . . . . See 4.6.47

n. Vibration failure . . . . . See 4.6 48

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o. Mechanical shock . . . . . See 4.6.49

p. Shock . . . . . See 4.6.50

q. Structure test . . . . . See 4.6.51

r. Pressurization . . . . . See 4.6.52

s. Pitch trim effect . . . . . See 4.6.55

t. Pitch trim torque . . . . . See 4.6.10.1.

4.5.2.2.1 At the option of the qualifying activity, the sampling plan B tests may be divided into two groups as follows, to be conducted on two sets of three instruments each if a need exists to expedite testing. Each indicator shall meet all individual tests after completing the group of specific tests to which submitted. At the option of the qualifying activity, group II tests may be conducted on empty cases, where testing permits, which have all external parts installed or attached as on a completed indicator. If this is allowed, the cases, parts and sealing operation shall be from current production.

Group I -	Sampling plan A tests	See 4.5.2 1
	Radio noise interference	See 4.6.37
	High temperature exposure	See 4.6.38
	Temperature shock	See 4.6.39
	Temperature altitude (cycling)	See 4.6.40
	Acceleration (nonoperating)	See 4 6.47
	Vibration failure	See 4.6 48
	Shock	See 4.6.50
	Mechanical shock	See 4.6.49
	Structure test	See 4.6.51
	Pressurization	See 4.6.52
	Pitch trim effect	See 4.6 55
	Pitch trim torque	See 4.6 10.1
Group II -	Sealing thermal shock	See 4 6 41
	Dust	See 4 6.42
	Humidity	See 4.6.43
	Fungus	See 4.6.44
	Salt-fog	See 4.6.45
	Rain	See 4.6.46.

4.5.2.3 Rejection and retest. When one item selected from a production run fails to meet the specification, items still on hand or later produced shall not be accepted until the extent and cause of failure are determined. After corrections have been made, all necessary tests shall be repeated.

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4.5.2.3.1 Individual tests may continue. For operational and production reasons, individual tests may be continued pending investigation of a sampling test failure. But final acceptance of indicators on hand or later produced shall not be made until it is determined that the indicators meet all the requirements of the specification.

4.5.2.4 Sampling plan C. Unless otherwise specified (see 6.2), 6 indicators shall be selected from the first 30 items on the contract or order and subjected to the reliability test specified in 4.6.54. Accept-reject criteria shall be as specified therein.

4.5.3 Defects in items already accepted. The investigation of a test failure could indicate that defects may exist in indicators already accepted. If so, the contractor shall fully advise the qualifying activity of all defects likely to be found and methods of correcting them.

4.6 Test methods

4.6 1 Examination of product. The indicator shall be inspected to determine compliance with the requirements specified herein with respect to materials, workmanship, marking, and dimensions The indicator shall be placed in the face-down position, shaken by hand, and inspected in this position to determine that no loose hardware or foreign material is present. Evidence of any such material shall be cause for rejection.

4.6 2 Early failure detection test. A 25-hour (indicator operating time) early failure detection test shall be performed on each indicator with amplifier attached. The test shall be conducted after the indicator has been calibrated and hermetically sealed All individual performance tests shall have been met prior to initiation of this test. Upon completion of the early failure detection test, the indicator shall be subjected to the individual tests. The following conditions shall exist during this test

a. Temperature cycling, on-off cycling and vibration shall be as specified in 4 6.53

b. During operating periods, indicator operation and observation thereof shall be as specified in 4.6.53 with the following exceptions·

(1) The step inputs to the vertical and horizontal pointers to drive them from view shall be 13.5 ma dc. The step inputs to drive the pointers to their extreme in-view positions shall be 5 ma dc.

(2) Step inputs to the displacement pointer to drive it to its out of view extreme position shall be 1,000 µa and the step inputs to drive it to its extreme in-view position shall be 500 µa

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(3) The pointer warning flags shall be driven from view by a step input of 380 µa dc after each de-energized period of the flag.

(4) The red lighting circuit shall be energized for 50 percent of the operating time, and the white lighting circuit shall be energized for 50 percent of the operating time.

4.6.3 Starting. The indicator shall be properly connected When power is applied, the warning flag shall disappear and the indicator shall be ready for immediate operation. The reading on the elapsed time indicator shall be recorded. At least ten other readings shall be made throughout the testing program. At least three readings shall be made during sampling plan A testing and seven readings during sampling plan B testing. At least five of the seven readings shall be made during the reliability test

4.6.4 Pitch and bank zero. The standard test transmitters in the pitch, roll, and azimuth axes shall be set to the 180° dial reading Power shall be applied and the indicator shall follow up to the zero indications in pitch and roll within ±0.5° from any random indication in the pitch and roll axis and within ±0.5° of a south heading This test shall be repeated for any initial indication at the discretion of the qualifying activity Pitch, roll, and azimuth indications shall be adjusted to zero by means of the zeroing adjustments specified in 3.10.6 1 1. This test shall be repeated with the standard test transmitters set to the 0° dial reading The pitch, roll and azimuth axes shall follow up to the zero indication (North heading for azimuth) within ±0.5°. These tolerances shall be met without further adjustment of the pitch, roll or azimuth trim adjustments.

* 4.6.5 Sensitivity test. The test transmitter for the roll axis shall be moved clockwise 0.25°. There shall be a perceptible movement of the bank index. The test transmitter for the roll axis shall then be returned 0.25° to its original position There shall be a corresponding perceptible movement of the indicator bank index. This test shall be repeated, except the initial transmitter movement shall be in the counterclockwise direction The above tests shall be repeated with the azimuth and pitch test transmitters and there shall be corresponding perceptible movements of the horizon line or azimuth axis, as applicable, for each separate movement of the test transmitter The test shall be repeated in the pitch axis with the pitch trim adjustment to its extreme positions, except each movement of the pitch test transmitter shall be 0.5°.

4.6.6 Followup rate. The indicator shall be properly connected to a remote signal source or to standard test transmitters. The followup rates in pitch, roll, and azimuth shall be determined. The rate in the roll and azimuth axes shall be not less than 300° per second. The rate in the pitch axis shall be not less than 90° per second with the pitch trim at zero and when adjusted to each of its extreme positions.

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4.6.7 Followup accuracy. The standard test transmitter in the roll axis shall be displaced to at least 10 random positions other than the 180° dial reading. The indicator shall position itself with the rotor of the standard test transmitter within ±0.5° for each position within ±30° indication and within ±1° at all other indications. The standard test transmitter in the pitch axis shall be displaced to at least 10 random positions other than the 180° dial reading. The indicator shall position itself with the standard test transmitter within ±0.5° for each position within ±30° indication and within ±1° for all other indications. The standard transmitter in the azimuth axis shall be displaced to 12 positions in increments of 30° from the 0° reading. The azimuth indication shall position itself with the standard test transmitter to within ±1° for the position chosen.

NOTE. Within ±30° indication is defined to be within ±30° of a 0° or a 180° test transmitter setting.

4.6.8 Followup operation. Suitable test equipment shall be connected to the appropriate test pins of the indicator to observe and record any hunting, jumping, lag, or overshooting of the indication when the indicator is made to follow the standard transmitter at various constant rates in pitch, roll, and azimuth axes. This shall be possible with the indicator sealed Hunting, jumping, overshoot and lag data shall be recorded each time the individual tests are performed. Recording equipment and procedures shall be subject to the approval of the procuring activity.

4.6.8.1 Indication lag. Indication shall not lag behind the shaft of the remote input transmitting synchro by more than the amount shown in the curve on figure 12. The indicator shall be tested at three different rates (including the maximum rate specified in 4 6.6) for each servo. The servos may be checked at any rate within the maximum rate at the discretion of the qualifying activity A 25 percent increase in indication lag will be permitted during temperature extremes and low voltage tests. No trim signals (pitch, roll, or azimuth) shall be applied when conducting lag tests

* 4.6 8.2 Hunting and jumping There shall be no noticeable hunting or jumping when the standard transmitter shaft is not rotating The manufacturer shall conduct a survey on the indicator which shall be operated over the full range of operating speeds. The testing rates chosen for hunting and jumping shall be those that produce the most erratic motion and shall include the following

a. Pitch Axis 1.5° per second, 20° per second, 90° per second

b. Roll Axis 1.5° per second, 20° per second, 90° per second, 300° per second

c. Azimuth Axis 1.5° per second, 20° per second, 90° per second, 300° per second

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The indicators shall be tested at the above rates separately in the cw and ccw directions. In addition, all three axes shall be subjected to a slew rate of 1.5° per second simultaneously in each direction. All three axes shall be recorded simultaneously. Evidence of hunting and jumping shall be checked at each rate as specified above. Each test at 1.5° per second shall be at least 4 minutes in duration. Tests shall be performed with zero trim The peak-to-peak amplitude of the hunting and jumping shall not exceed 0.125° in any axis for shaft speeds up to 90° per second in pitch or 300° per second in roll and azimuth. The hunting and jumping tests shall be repeated in the pitch axis with the pitch trim knob adjusted to deflect the horizon line up to 10°. Hunting and jumping shall not exceed 0.2°.

4.6.8.3 Overshooting. The pitch, roll and azimuth axes shall each be subjected to a step input equal to the maximum allowable lag The indicator shall stabilize at the ordered position within 0.20 seconds without further overshooting or oscillation.

4.6.8.4 0° to 180° switching. The indicator shall be suitably connected to the test transmitters so that any two axes can be concurrently and instantaneously switched from 0° transmitter signal to a 180° transmitter signal and vice versa. In each of the following conditions, the test transmitter signals to the indicator shall be switched from the initial to final condition and the applicable axes of the indicator shall follow up by rotating 180°. If the indicator fails to respond at any of the specified conditions, it shall be rejected. This test shall be repeated switching from the final to the initial condition for each case.

Initial Condition	Final Condition
a. Pitch - 0° Roll - 0°	Pitch - 180° Roll - 180°
b. Pitch - 180° Azimuth - 0°	Pitch - 0° Azimuth - 180°
c. Roll - 0° Azimuth - 180°	Roll - 180° Azimuth - 0°

4.6.9 Gimbal freedom. The indicator shall be properly connected and the remote signal source or standard test transmitter moved 360° in roll, pitch, and azimuth. The indicator shall follow smoothly without sticking and overshooting or hunting greater in magnitude than that specified in 4 6.8. There shall be no disturbance of any of the meter movements as a result of this test.

4.6.10 Pitch trim. The pitch trim knob shall be adjusted to its limit and the deflection of the horizon line observed. The deflection shall be from 5° to 10° down and from 10° to 20° up from the level flight indications.

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4.6.10.1 Pitch trim torque. The torque required to rotate the pitch trim knob shall be measured throughout the range of settable indications in both the cw and ccw directions and shall be from 1 inch-ounce to 8 inch-ounces. A minimum of five readings shall be taken.

4.6.10.2 Pitch trim fade. The pitch trim knob shall be rotated to the cw stop. The pitch test-transmitter rotor shall be rotated from level flight (0°) to 80° pitchup (climb). There shall be no discontinuity in the pitch indication. At the 80° climb, the pitch trim knob shall have no effect on the pitch indication. With the test transmitter at 65° pitchup (climb), the pitch trim range shall be perceptible but less than that exhibited in the level flight condition (0°). This test shall be repeated in the ccw and dive directions and then in inverted flight (180°) instead of level flight (0°). The pitch trim sensing shall be reversed in the inverted flight condition, i.e., cw rotation shall cause the horizon line to move down.

4.6.10.3 Pitch trim zero. With the test transmitter set at 180°, the pitch trim knob shall be set to zero (center of the dot) and the indicator shall follow up to within ±0.5° of zero pitch indication.

4.6.11 Zeroing adjustments. The indicator shall be properly connected to the standard test transmitters with each set to electrical zero. The roll zeroing adjustment shall be moved to its extremes, and the deflection of the bank index shall be not less than ±3° from the zero position. The azimuth zeroing adjustment shall be moved to its extremes, and the azimuth deflection shall be not less than 3°. The pitch test transmitter shall be rotated 3° in climb and then 3° in dive. In each case, the pitch trim knob shall be loosened on its shaft and the shaft rotated until the indication is zeroed. The pitch trim knob shall be tightened again to the shaft at the zero position. In each case, the requirements of 4.6.10 shall be met. This test shall be performed only during the initial individual tests and during the final individual tests that are conducted as a part of the reliability test.

4.6.12 Sealing. The case of the indicator proper shall be tested for leaks by means of a mass-spectrometer type helium leak detector. The maximum leak rate shall not permit more than 10 percent contamination of the filling medium after 1,000 hours at a pressure differential of 12 to 14.7 psi.

4.6.13 Power off flag. The indicator shall be properly connected, but no power applied. Power shall then be applied, and the flag shall disappear. The power shall be disconnected, and the flag shall immediately appear. Power shall be applied, and the jumper from pin J to pin A shall be disconnected. The flag shall appear.

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4.6.14 Slip indicator sensitivity. With the dial vertical, the indicator shall be rotated about the roll axis to the right until the ball is just short of its limit. The angle of rotation shall be 7° ±2°. The test shall be repeated rotating the instrument to the left. The same tolerance shall apply. The ball shall not stick at the high end of the tube.

* 4.6.15 Slip indicator zero position. The indicator shall be positioned horizontally with reference to the two top mounting holes of the mounting bezel and gently tapped. The ball shall rest at the zero mark of the slip indicator within ±1/32 inch.

4.6.16 Slip indicator friction. The inclinometer ball shall roll smoothly when the indicator is tipped to either side in the vertical plane of the dial at a rate of 2° +0.1° -0.0° per second.

4.6.17 Slip indicator visibility. The indicator shall be tipped so that the ball will be at rest at either end of the tube. Not less than two-thirds of the ball shall be visible when the ball is viewed from the position specified in 3.10.8.2.

4.6.18 Slip indicator filling. The indicator shall be tipped so that all air in the tube will be trapped in the expansion chamber. The indicator shall then be placed in the level position and rotated around its longitudinal axis to a bank angle of 12° with the expansion end of the tube low. During this rotation, no part of the air bubble shall be visible when viewed from the position specified in 3.10.8.2.

4.6.19 Response. The response of the vertical, horizontal, displacement, and rate-of-turn pointers shall be measured and shall be in accordance with 3.10.9.1.2, 3.10.9.3.2, 3.10.9.4.2, and 3.10.10.2, respectively. A step input equivalent to that required to drive each pointer out of view shall be applied ten consecutive times. No damage or sticking shall result.

4.6.19.1 Response time. The response time of the pointers shall be measured and shall be in accordance with 3.10.9.1.5, 3.10.9.3.5, 3.10.9.4.5, and 3.10.10.5, respectively. For this test, the instrument shall be shunted externally with a resistance of 350 ohms to allow for the shunting effect of another indicator in parallel with the usual source. The voltage source shall be steady and shall be applied to the above circuit through at least 50,000 ohms. At least 5 seconds shall be allowed in order to establish calibration of the 100-percent mark.

4.6.20 Pointer and flag balance. With the indicator in the normal upright position and the pointer mechanism de-energized, the vertical and horizontal pointers shall each be adjusted by means of the zero correctors to bisect the center of the indication. The vertical, horizontal, and rate-of-turn pointers

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shall not deflect more than 1/64 inch and the displacement pointer shall not deflect more than 1/32 inch as the indicator is rotated 360° in roll and pitch. The OFF flag and pointer alarm flags shall not move off their visible position stops when the indicator is rotated 360° about the roll axis in 90° increments.

* 4.6.20 1 Power shall be applied to the flag circuits to drive all flags out of view. The indicator shall be subjected to 10g's acceleration along its normal vertical axis. The pointers shall not vary by more than 5/64 inch from their zero position nor shall the warning flags or off flag come into view While under acceleration the warning flags and off flag shall be de-energized and they shall come into view. The warning flags and off flag shall then be energized and they shall go out of view. This test shall be repeated with 10g acceleration applied to the longitudinal axis and 6g acceleration applied to the lateral axis.

4.6.21 Pointer alarm flags. The response of the pointer alarm flags shall be measured and shall be in accordance with 3.10.9.2.2.

4 6.22 Lighting (brightness and color)

4.6.22 1 The following tests shall be performed as individual tests during qualification and sampling plan A, B, and C tests:

4.6.22.1.1 Light The indicator shall be subjected to the light tests (brightness, col ', and stray light) as specified in MIL-L-27160 and MIL-L-25467.

4.6.22.1 Subarea comparisons. The areas specified in 3.14 shall be measured. The airp! ne may be divided into subareas, such as each wing, center dot, and the white areas in between. The average brightness of each area shall be compared with the maximum brightness on that portion of the gray sphere adjacent to that area. The maximum brightness of the gray sphere shall not exceed 0.60 times the minimum brightness of the individual sections of the airplane specified above.

* 4.6 22 2 The following tests shall be conducted only during the individual production quality conformance testing:

4.6 22.2.1 Visual comparison. This test shall be conducted by qualified personnel only after they have been tested for normal color vision and have spent at least 20 minutes in the dark to become dark adapted. The prime standard and the indicator under test shall be placed side-by-side at eye level in completely dark surroundings and viewed at a distance of approximately 2.5 feet in a sighting cone 30° from normal to the surface of the cover glass. All indicator markings, pointers, flags, et cetera, shall be visible when viewed from any position within the sighting cone.

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4.6.22.2.2 Brightness. With 2.50 ±0.01V applied to the indicator under test and 2.40 and 2.60V applied consecutively to the lighting terminals of the prime standard, like areas of the pointers, command bars, alarm flags, sphere, numerals, indicia, markings, et cetera, of the indicator under test and the prime standard shall be compared for brightness. The two instruments shall also be compared for lighting uniformity. The brightness of the production unit at 2.50V shall fall within the brightness settings of the prime standard when set at 2.40 and 2.60V. If the brightness is dissimilar in any individual area or the lighting of the indicator and prime standard is not uniform, the questionable areas of the production units shall be measured with a photometer in accordance with MIL-L-25467 or MIL-L-27160, as applicable. If the production units fail this test, they shall be rejected and returned for rework. Qualification units shall not be tested by use of the prime standard.

4.6.22.2.3 Color. The color comparison test shall be conducted as specified in 4.6.22.1.1, except 4.50V shall be applied to both the production indicator and the prime standard. The two indicators shall be identical in color.

4.6.23 Fogging. The indicator shall be energized for a minimum of 1 hour in a 71° ±2°C controlled ambient. After this period and while at the specified temperature, an ice cube shall be rubbed on the indicator glass face for 1 to 2 minutes. The glass shall be wiped dry (do not use compressed air) and the indicator inspected for evidence of water or oil fog. Evidence of fogging shall be cause for rejection.

4.6.24 Bank angle command pointer adjustment (ARU-31/A only). The pointer shall be adjusted through its entire range and it shall travel 62° +2°/-0° in the clockwise and counterclockwise directions from the zero roll position. Three ±1.0 rotations shall be required to move the pointer from cw to ccw stop. The bank angle command pointer knob shall be rotated beyond its cw and ccw stops without further movement of the pointer or damage to the pointer mechanism.

4.6.24.1 Bank angle command pointer torque. The torque required to rotate the bank angle command pointer knob shall be measured throughout the range of settable indications in both cw and ccw directions. A minimum of 10 readings shall be taken. The torque shall be from 2 to 12 inch-ounces.

4.6.25 Power consumption. The ac power consumption of the indicator shall be measured with suitable meters and shall not exceed 18 va. The power factor shall be maintained between 0.7 and 1.00. The power consumption and power factor shall be measured with the pitch servo operating at 90° per second and the roll and the azimuth servo operating at 300° per second. Power consumption of each lighting circuit shall not exceed 10 va.

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4.6.26 Low voltage. The indicator shall be properly connected. The ac voltage shall be reduced to 103V and 380 Hz. The indicator shall meet all tests except 4.6.12, 4.6.14, 4.6.15, 4.6.16, 4.6.17, 4.6.17, 4.6.19, 4.6.20, 4.6.21, 4.6.22, 4.6.23, and 4.6.24. The tests shall be repeated for all combinations of 380 and 420 Hz and 103 and 127V.

4.6.27 Magnetic property. This test shall be conducted first with no power applied to the indicator and then repeated with the indicator operating on rated power. The indicator shall be held in various positions magnetically east or west of and not more than 12 inches from the center of a free magnet approximately 1-1/2 inches long in a magnetic field with a horizontal intensity of 0.18 ±0.01 oersted. The maximum deflection of the magnetic shall not exceed 5°. An aircraft compass with the compensating magnetics removed may be used as the free magnetic for this test

* 4.6.28 Acceleration (operating). The indicator shall be properly connected and power shall be applied to the attitude circuit. The indicator shall be accelerated with 10g along its normal vertical axis. The attitude display shall not vary by more than 1° in pitch or roll from the initial indication and the pointers shall not vary by more than 5/64 inch from the zero position. The bank angle command pointer shall not vary by more than the width of a 2° bank index mark (ARU-31/A only). This test shall be conducted with the attitude set at zero pitch and roll and with each axis displaced 90°.

4.6.29 Insulation resistance. A potential of 500V dc shall be applied for 10 seconds between isolated pins of the connector (Scintilla 10-150631-32P, or equal) and between pins of the connector and the indicator. The resistance shall be not less than 10 megohms.

4.6.30 Vibration error. The indicator shall be properly connected, mounted, and subjected to a vibration test in accordance with MIL-STD-810, Method 514, procedure I, part 1, figure 514.1-2, curve C, except the range shall be from 5 to 50 cps, the acceleration level shall be ±1g between 23 and 50 cps, the cycling time from 5 to 50 to 5 cps shall not be less than 2 minutes in each axis and resonance search and dwell are not required.

4 6 30.1 Roll, pitch, and azimuth, and pointers. The maximum amplitude of the sphere in oscillation in pitch, roll, and azimuth throughout the frequency range shall not exceed 1/2° from its previbration state. The maximum total amplitude of the horizontal, vertical, displacement, and rate-of-turn pointers (hereinafter referred to as the pointers) oscillation throughout the frequency range shall not exceed one-half the width of the respective pointer.

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* 4.6.30.2 Roll, pitch, and azimuth, and pointer variation. Throughout the frequency range, the maximum variation during vibration of the sphere in pitch, roll, or azimuth or of the pointers from their original indication before starting the vibration shall be noted and shall not exceed 1/2° in pitch, roll, or azimuth, nor one-half the width of the respective pointer. The bank angle command pointer shall not vary by more than the width of a 2° bank angle index mark (ARU-31/A only).

4.6.30.3 Oscillation and variation tests may be conducted at any indication.

4.6.30.4 Slip indicator accuracy. The slip indicator reading shall not differ at any time more than 1/16 inch from the reading obtained when the indicator is at rest.

4.6.31 Horizon line and bank index null position shift. The indicator shall be properly connected to standard test transmitters in the pitch, bank, and azimuth axes. The transmitters shall be set to drive the indicators in all three axes to the zero position. The input voltage and frequency shall be varied within the following limits and combinations:

Frequency	380 to 420 Hz
AC voltage	103 to 127V.

The horizon line shall not shift more than ±1/16 inch from normal null position and bank index shall not shift more than 0.25° from normal null position.

4.6.32 Low temperature operation. With no power applied, the indicator shall be properly connected and placed in a chamber at a temperature of -54° ±2°C for 4 hours. At the end of this period and with the indicator at the reduced temperature, power shall be applied for 10 minutes after which the indicator shall meet the individual tests and the test specified in 4.6.35 except:

a. The displacement time for the inclinometer ball shall be 7 seconds maximum

b. The meter movements shall be checked only for operation without evidence of sticking or binding, and to ascertain that they have not shifted by more than ±1/64 inch from their null settings at room temperature conditions

c. The tests specified in 4.6.12, 4.6.19, 4.6.20.1, and 4.6.22 need not be conducted until the indicator has returned to room temperature

d. The tests specified in 4.6.23 need not be conducted

e. The tests specified in 4.6.8.1 and 4.6.8.2 shall be limited to 200°/sec maximum rates for the roll and azimuth axes.

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4.6.33 High temperature operation. The indicator shall be properly connected and placed in a chamber at a temperature of 71° ±2°C and operated with power applied for 4 hours. At the end of this period and while at the high temperature, the indicator shall be subjected to and shall meet the individual tests except

a. The meter movements shall be checked only for operation without sticking or binding, and to ascertain that they have not shifted more than ±1/64 inch from their null settings under room temperature conditions

b The tests specified in 4.6.12, 4.6.19, 4.6.20.1, and 4.6.22 need not be conducted until the indicator has returned to room temperature.

4.6.34 Slip indicator leakage. Prior to the high temperature test, the size of the air bubble in the tube shall be determined with the indicator at room temperature. Following the high temperature test, the size of the air bubble shall again be determined with the indicator at room temperature. There shall be no appreciable change in size.

4.6.35 Slip indicator damping, room temperature. The plane of the face of the indicator shall be placed in a vertical position with the upper flange inclined to the horizontal at an angle of 12°. The indicator shall be tipped suddenly through a vertical angle of 24° so that the ball will roll to the opposite end of the tube. The time required for the ball to roll from the zero mark of the indicator to the end of the tube shall be between 0.15 and 0.50 second.

4.6.36 Lighting circuit test. The indicator shall be subjected to all the tests as specified in MIL-L-27160 and MIL-L-25467, except the Life test shall be performed in conjunction with the Reliability tests.

4 6.37 Radio noise interference. The radio noise interference test shall be conducted in accordance with MIL-I-6181. During this test the pitch, roll, and azimuth axes shall be oscillated ±15° about zero indication at a rate between 1.5° per second to 5° per second Any out-of-specification condition shall be investigated and necessary action taken before retesting is accomplished.

4 6 37.1 ARU-11/A indicator. In addition to meeting the test specified in 4.6.37, the ARU-11/A indicator shall meet the following:

4.6.37.1.1 Radio frequency conducted susceptibility. The test signal shall be increased from 100,000 microvolts to (a) 3V in the frequency range from 0.150 to 30 megacycles and (b) to 1V from 30 to 400 megacycles.

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4.6.37.1.2 Radio frequency radiated susceptibility. The test signal shall be increased from 100,000 microvolts to (a) 10V per meter in the frequency range from 0.150 to 35 megacycles and (b) to 5V per meter from 35 to 1,000 megacycles. Measurements above 1,000 megacycles are not required.

4.6.37.1.3 Switching-transient susceptibility. The indicator shall operate without malfunction, change in indication, or degradation of performance when exposed to a switching-transient electromagnetic field generated by an MS25271 relay and wired for continuous operation as shown on figure 13. Suppression components or diodes shall not be used. The relay circuit shall be unshielded wire taped to or in parallel with the equipment input power leads, signal leads, and interconnecting leads and looped about the black boxes that comprise the test samples.

4.6.38 High temperature exposure. The indicator shall be properly connected and subjected to a high temperature exposure test in accordance with MIL-STD-810, Method 501, procedure I, for 48 hours with no power applied. At the end of this period and while still at the high temperature, the indicator shall meet the test specified in 4.6.9. All meter movements shall operate without binding or sticking and shall be checked to ascertain that they have not shifted more than ±1/64 inch from their null settings under room conditions. At the conclusion of this test, with the indicator at room temperature, the indicator shall be checked for satisfactory operation. Satisfactory operation is considered as proper operation of all servos, meters, flags, without sticking or binding at any indication. There shall be no slip indicator leakage.

4.6.39 Temperature shock. The indicator shall be subjected to a temperature shock test in accordance with MIL-STD-810, Method 503, after which it shall meet the individual tests with the exception of early failure detection.

4.6.40 Temperature-altitude (cycling). The indicator shall be properly connected and subjected to a temperature-altitude (cycling) test in accordance with MIL-STD-810, Method 504, class 2 equipment, except that step 7 shall be conducted at 71°C and step 10 shall not be conducted. Satisfactory operation is considered as the proper operation of all servos, meters, flags, and pointers without sticking or binding at any indication. At the conclusion of this test and after the indicator has returned to room temperature, it shall meet the individual tests with the exception of the early failure detection. There shall be no slip indicator leakage.

4.6.41 Sealing thermal shock. The indicator shall be immersed alternately in tap water maintained at 85° ±5°C and 5° ±4°C for a total of 8 cycles. The length of time for each bath immersion shall be 30 minutes. Not more than 5 seconds shall elapse between bath immersions. No damage to the hermetic seal shall result due to this test. Following the immersions, the indicator proper shall be tested for leaks by means of a mass spectrometer type of helium leak detector. The initial maximum detected leak rate at a differential pressure of 1 atmosphere shall not permit more than 10 percent loss of the total filling after 1,000 hours.

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4.6.42 Dust. The dust test shall be conducted in accordance with MIL-STD-810, Method 510. At the end of this test, the indicator shall be removed from the test chamber and allowed to cool at room temperature. The indicator shall then pass the tests specified in 4.6.9 and 4.6.23. All adjustments (pitch trim, bank angle command pointer knob, pointer adjustment screws, and zeroing adjustments) shall operate properly over their complete range without excessive stickiness, looseness, or binding.

4.6.43 Humidity. The humidity test shall be conducted in accordance with MIL-STD-810, Method 507, procedure I. At the conclusion of this test, the indicator shall pass the tests specified in 4.6.9 and 4.6.23. All adjustments (pitch trim, band angle command pointer knob, pointer adjustment screws, and zeroing adjustments) shall operate properly over their complete range without excessive stickiness, looseness, or binding. There shall be no evidence of corrosion or rust of internal or external parts as a result of this test.

4.6.44 Fungus. The indicator, with amplifier attached, shall be subjected to a fungus test in accordance with MIL-STD-810, Method 508 after which the indicator shall pass the tests specified in 4.6.9 and 4.6.23. All adjustments (pitch trim, bank angle command pointer knob, pointer adjustment screws, and zeroing adjustments) shall operate properly over their complete range without excessive stickiness, looseness, or binding. There shall be no evidence of corrosion or rust of internal or external parts of any other deterioration that would adversely affect subsequent operation at a result of this test.

4.6.45 Salt fog. The salt fog test shall be conducted in accordance with MIL-STD 810, Method 509 for 50 hours. At the end of this period, the indicator shall pass the tests specified in 4.6.9 and 4.6.23. All adjustments (pitch trim, bank angle command pointer knob, pointer adjustment screws, and zeroing adjustments) shall operate properly over their complete range without excessive stickiness, looseness, or binding. There shall be no damage as a result of this test, and the tolerances specified under the individual tests shall apply.

4.6.46 Rain. The indicator shall be subjected to a rain test in accordance with MIL-STD-810, Method 506, except it shall be mounted in a rain test chamber for 1/2 hour with the dial in a vertical plane and for 1/2 hour with the dial in a horizontal plane (dial face upward). Upon removal from the chamber, the indicator shall pass the tests specified in 4.6.8 and 4.6.23. All adjustments (pitch trim, bank angle command pointer knob, pointer adjustment screws, and zeroing adjustments) shall operate properly over their complete range without excessive stickiness, looseness, or binding.

4.6.47 Acceleration (nonoperating). The indicator, not operating, shall be mounted in its normal position and accelerated first along its vertical axis and then along each of two axes that are perpendicular to the vertical axis and to each other. The indicator shall be subjected to an acceleration of 20g in each of the above axes for 1 minute after which it shall pass the individual tests. No damage shall result from this test.

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4.6.48 Vibration failure. The indicator shall be subjected to a vibration test over the frequency range from 5 to 50 Hz in accordance with MIL-STD-810, Method 514, procedure I, part 1, figure 514.1-2, curve C from 5 to 33 Hz and curve B from 33 to 50 Hz. The amplifier shall be tested over the frequency range from 5 to 500 Hz in accordance with procedure I, part 1, figure 514.1-2, curve D. At the conclusion of this test, the indicator shall be properly connected and shall pass the individual tests. No looseness in the mechanism nor damage to any part of the indicator shall result from this test.

4.6.49 Mechanical shock. The indicator shall be subjected to a shock test in accordance with MIL-STD-810, Method 516, procedure I, figure 516.1-2. Power shall be applied to the indicator and the attitude sphere shall be positioned to a 0° pitch and roll indication during this test. The indicator shall then pass the individual tests.

4.6.50 Shock. The indicator shall be subjected to a shock test in accordance with MIL-STD-810, Method 516, procedure II. At the end of this test, the indicator shall pass the individual tests.

4.6.51 Structure test. A structure test shall be conducted to ascertain that neither the indicator electrical connector nor the indicator case is damaged or deformed in any way when a total force up to 60 pounds is applied to the connector along the longitudinal axis of the instrument while inserting the indicator into a stationary mount when installing it is the aircraft. The testing apparatus and procedure to be utilized shall be approved by the qualifying activity prior to conducting this test.

4.6.52 Pressurization. The indicator shall be placed in a chamber at an absolute pressure of 28 psi and a temperature of 25°C for 4 hours. During this period, no deformation nor other damage to the indicator shall occur. At the conclusion of this test, the indicator shall be closely examined for evidence of deformation. Evidence of deformation, damage, or any unsatisfactory operation shall be cause for rejection. After this test, the indicator shall pass the leak-rate test portion of the sealing thermal shock test specified in 4.6.41.

4.6.53 Qualification reliability. The six test samples specified in 4.4.1 shall be subjected to a reliability test in accordance with MIL-STD-781, test level F except vibration shall be 1.0g ±10 percent, and test plan I except as shown on figure 14. This test shall demonstrate a minimum acceptable MTBF of 1,000 hours at a 90 percent confidence level. Failure categories, analysis, repair, and reporting shall also be in accordance with MIL-STD-781.

4.6.53.1 Duty cycle. The following conditions shall exist during the reliability test:

a. Power shall be applied after the indicators have stabilized at low temperature

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b. Three of the indicators shall have 5V applied to the red lighting circuit and three indicators shall have 5V applied to the white lighting circuit for the test duration. The lighting circuits shall be de-energized for a period of 0.5 to 3 minutes once each operating hour.

c. Ten minutes after power application, indicator operation shall begin. The duty cycle shall consist of the following modes of operation:

(1) Pitch, roll, and azimuth axes shall be continuously oscillated ±30° from the zero degree indication at a rate of 1.5°/second for 80 percent of each operating period.

(2) All three axes shall be driven through their full range in the clockwise direction at a constant rate of 5°/second for 5 percent of each operating period.

(3) All three axes shall be driven through their full range in the counterclockwise direction at a constant rate of 5°/second for 5 percent of each operating period.

(4) All three axes shall remain excited, but shall not be driven for 10 percent of each operating period. This condition shall be present while the indicator is at high temperature.

(5) The vertical, horizontal, displacement, and rate of turn pointers shall be driven through their complete operating ranges (not out of view) six times per minute. Each pointer shall move smoothly over its entire range. Once each operating hour, all pointers shall be subjected to a step input equivalent to that necessary to drive them to one extreme position and then to the opposite extreme position. No damage or stickiness shall occur.

(6) The power failure warning flag and pointer alarm flags shall be de-energized once every operating hour for a period of not more than 1 minute.

(7) The pitch trim knob shall be rotated to its extremes in both directions twice each operating hour.

(8) The bank command pointer knob (ARU-31/A only) shall be rotated through its operating range four times each operating hour. Once each operating hour, it shall be rotated to each stop and turned one additional turn against the stop.

d. Once each operating cycle, the indicators shall be observed for the first 15 minutes and once each hour thereafter to determine if any of the indicator functions are not operating properly. At least two of these observations shall be made during vibration periods.

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e. Prior to the test, at 500-hour intervals, and at the completion of testing, the indicator shall be subjected to the individual tests. In addition, at the completion of testing the indicators shall be subjected to the low voltage test, horizon line and bank index null position shift, and response time tests. Failures occurring during these tests shall be counted in the accept/reject decision.

4.6.54 Quality conformance reliability test. The six indicators specified in 4.5.2.4 shall be subjected to the reliability test specified in 4.6.53 with the following exceptions:

a. The accept/reject criteria shall be in accordance with MIL-STD-781, test plan III

b. If an accept decision cannot be made after accumulation of 9,940 total operating hours or if a reject decision is reached, testing shall be stopped. Further equipment deliveries shall not be made until corrective action has been taken and incorporated into all units. The effectiveness of the corrective action shall be demonstrated by testing as directed by the qualifying activity.

4.6.55 Pitch trim effect. Two indicators shall be connected in parallel to the same power source with both pitch synchro control transformers connected to the same standard test transmitter. The standard test transmitter shall be an Eclipse-Pioneer AY200-1, or equal, and shall be connected as specified in 3.10.1.3 and 3.10.2, except that a calibration dial need not be provided. The pitch transmitter stator leads shall also be connected to a third Eclipse-Pioneer AY500-5 synchro control transformer, or equal, having a 10,000-ohm resistance load connected across the rotor leads. The following test procedure shall apply:

a. With the pitch trim knob of each indicator set to a visual indication of zero pitch trim, the test transmitter rotor shall be positioned to produce a 0° ±5° pitch indication on the two indicators.

b. The rotor of the third synchro control transformer shall be positioned to produce null voltage across the control transformer rotor winding of 100 mv or less as measured by a vacuum tube voltmeter.

c. Rotation of the pitch trim knob of either indicator from the zero trim position to the extreme position in either direction shall not produce a change in the null voltage across the third control transformer rotor of more than ±20 mv from the initial null voltage reading. The pitch trim knob of the other indicator shall be maintained in the zero position during this test.

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d. This test shall be repeated with the transmitter rotated 180° ±5° from the original position.

4.6.55.1 The test specified in 4.6.55 shall be repeated using an Eclipse-Pioneer AY200-3 test transmitter, or equal. The same tests shall again be conducted using a Kearfoot CM01012-107 test transmitter, or equal.

4.6.56 Maintainability demonstration. Unless otherwise specified (see 6.2), the indicator shall be subjected to a maintainability demonstration in accordance with MIL-STD-471, test method 1A. A maintainability test plan in accordance with Section 4 of MIL-STD-471 shall be submitted and approved by the qualifying activity prior to initiating the maintainability demonstration test. Producers risk (α) and consumers risk (β) shall be 0.05 respectively. The mean maintenance time (H_0) shall not exceed 240 minutes not including the time to seal and unseal the unit. The maximum maintenance time (H_1) shall not exceed 360 minutes not including the time to seal and unseal the unit.

4.7 Inspection of preservation, packaging, packing, and marking. Inspection of preparation for delivery shall be conducted in accordance with the quality assurance provisions of MIL-STD-794 to insure compliance with the requirements of Section 5.

* 5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified. (See 6.2).

5.1.1 Level A. Indicators shall be preserved and packaged Method IC-1 in accordance with MIL-P-116 and MIL-STD-794, one each in unit containers conforming to PPP-B-636, domestic class.

5.1.2 Level C. Indicators shall be individually preserved and packaged in a manner that will provide protection against corrosion; deterioration or physical damage so that serviceability is assured during shipment from supply source to first receiving activity.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A. Indicators preserved and packaged as specified in 5.1.1 shall be packed in shipping boxes conforming to PPP-B-636, weather-resistant class, unless otherwise specified by the procuring activity. Insofar as practical, exterior shipping containers shall be of uniform shape and size of minimum tare, weight and cube.

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5.2.2 Level B. Indicators packaged as specified in 5.1.1 shall be packed in domestic class shipping containers conforming to PPP-B-636. Insofar as practicable, shipping containers shall be of uniform shape and size and of minimum cust and tare consistant with the protection required.

5.2.3 Level C. Indicators shall be packed in exterior type shipping containers in a manner that will ensure safe transportation at the lowest rate to the point of delivery and shall meet, as a minimum, the requirements of carrier and rules and regulations.

5.3 Marking of shipments. Interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129. The nomenclature shall be:

INDICATOR, ATTITUDE DIRECTOR(*)

(*) Applicable type designation to be entered by the contractor.

5.3.1 Reinspection date. The reinspection date markings shall be as specified by the qualifying activity (see 6.2).

6. NOTES

6.1 Intended use. The remote attitude-director indicators covered by this specification are intended for use in high-performance aircraft as part of an integrated panel. The indicators will be controlled by a vertical reference control providing transmitting synchro pitch, roll, and azimuth output signals and a flight-director computer providing director signals for the dc meter movements.

6.2 Ordering data. Procurement documents should specify:

a. Title, number, and date of this specification

b. Type of indicator required (see 1.2)

c. Data requirements (see 3.2)

d. When sampling plan B tests will not be conducted (see 4.5.2.2)

e. When sampling plan C tests will not be conducted (see 4.5.2.4)

f. When the maintainability demonstration test will not be conducted (see 4.6.56)

g. Levels of packaging and packing required and reinspection date marking (see Section 5).

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6.3 Definition

6.3.1 Hermetic seal. Hermetic sealing is the process by which an item is totally enclosed by a suitable metal structure or case and sealed airtight by fusion of metallic or ceramic materials. This includes the fusion of ceramic materials into a metallic support. Elastomeric or resinous materials, or combinations of these materials, may be used as a coverglass seal provided the specified leak rate requirements are met under all tests and environmental conditions specified herein. Elastomeric or resinous materials may be used in other areas provided prior approval is obtained from the qualifying activity. These materials should be in accordance with MIL-S-7502, type B-2, or MIL-A-8623, type II or type III. If these materials are used, the surfaces to be bonded must be cleaned in accordance with MIL-A-9067. A hermetic seal is not intended to include seals accomplished by gaskets or O-rings.

6.4 Qualification. With respect to products requiring qualification, awards will be made only for such products which are, at the time set for opening of bids, qualified for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order than they may be eligible to be awarded contracts or orders for the products cover by this specification. The activity responsible for the Qualified Products List is the Aeronautical Systems Division, Attn: ASD/ENFIF, Wright-Patterson Air Force Base, Ohio 45433, and information pertaining to qualification of products may be obtained from that activity.

6.5 International standardization agreement. Certain provisions (figure 1 and 3.19.1) of this specification are the subject of international standardization agreements STANAG 3435, ASCCAS 10/6, and STANAG 3329. When amendment, revision, or cancellation of this specification is proposed, the departmental custodians will inform their respective Departmental Standardization Offices so that appropriate action may be taken respecting the international agreements concerned.

6.6 Identification of changes. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the latest previous issue.

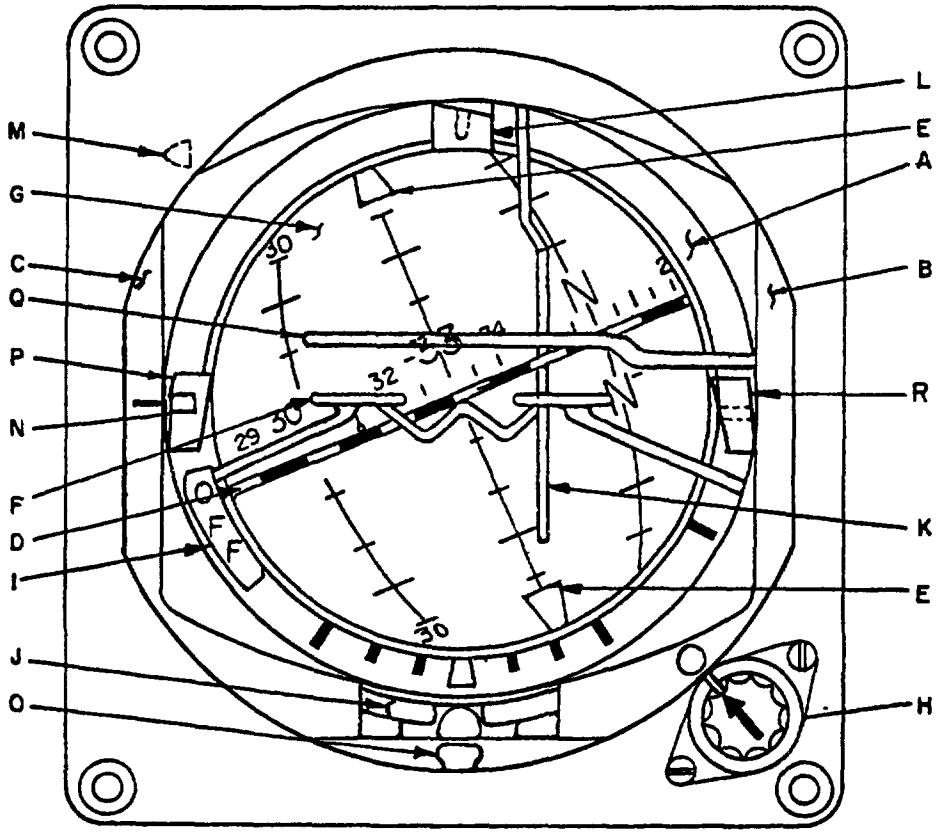
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Navy - AS

Review activity:
Air Force - 71

Preparing activity:
Air Force - 11

Project No. 6610-0368

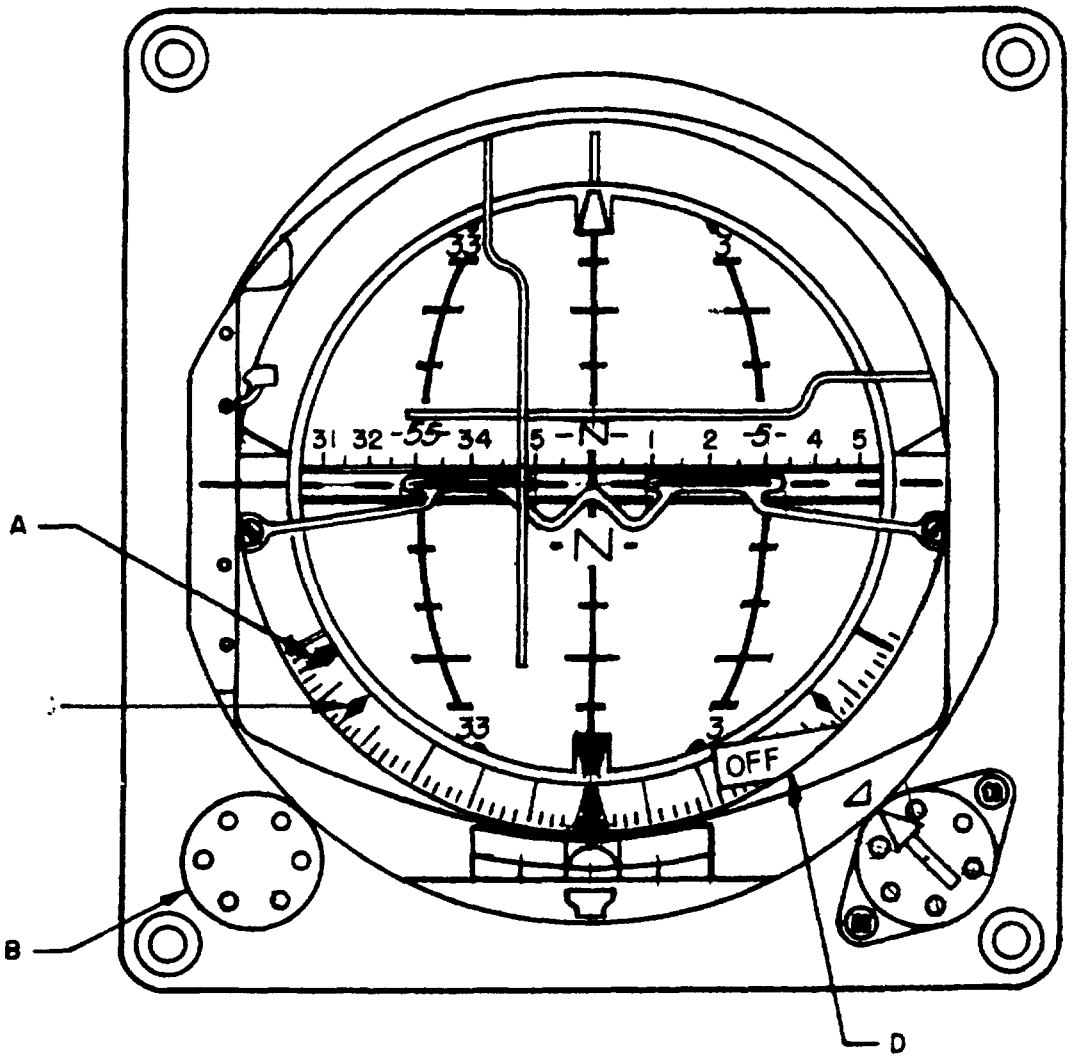
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A. Bank angle, dial or bank scale
B Dial mask and vertical pointer shield
C. Displacement pointer scale or glide slope deviation scale
D Mechanism mask
E. Bank angle indexes
F. Minitature airplane
G Sphere
H. Pitch trim knob
I. Power off flag
J. Slip indicator
K. Vertical pointer or bank steering bar
L Vertical pointer alarm flag or course warning flag
M. Displacement pointer out of view
N. Displacement pointer or glide slope indicator
O. Rate of turn
P. Displacement pointer alarm flag or glide slope warning flag
Q. Horizontal pointer or pitch steering bar
R. Horizontal pointer alarm flag (ARU-21/A only)

FIGURE 1. Indicator Display

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A. BANK ANGLE COMMAND POINTER
B. BANK ANGLE COMMAND POINTER ADJUSTMENT KNOB
C. 45° INDEX MARK
D. POWER OFF FLAG

FIGURE 2. ARU-31/A Precise Bank Attitude Director Indication

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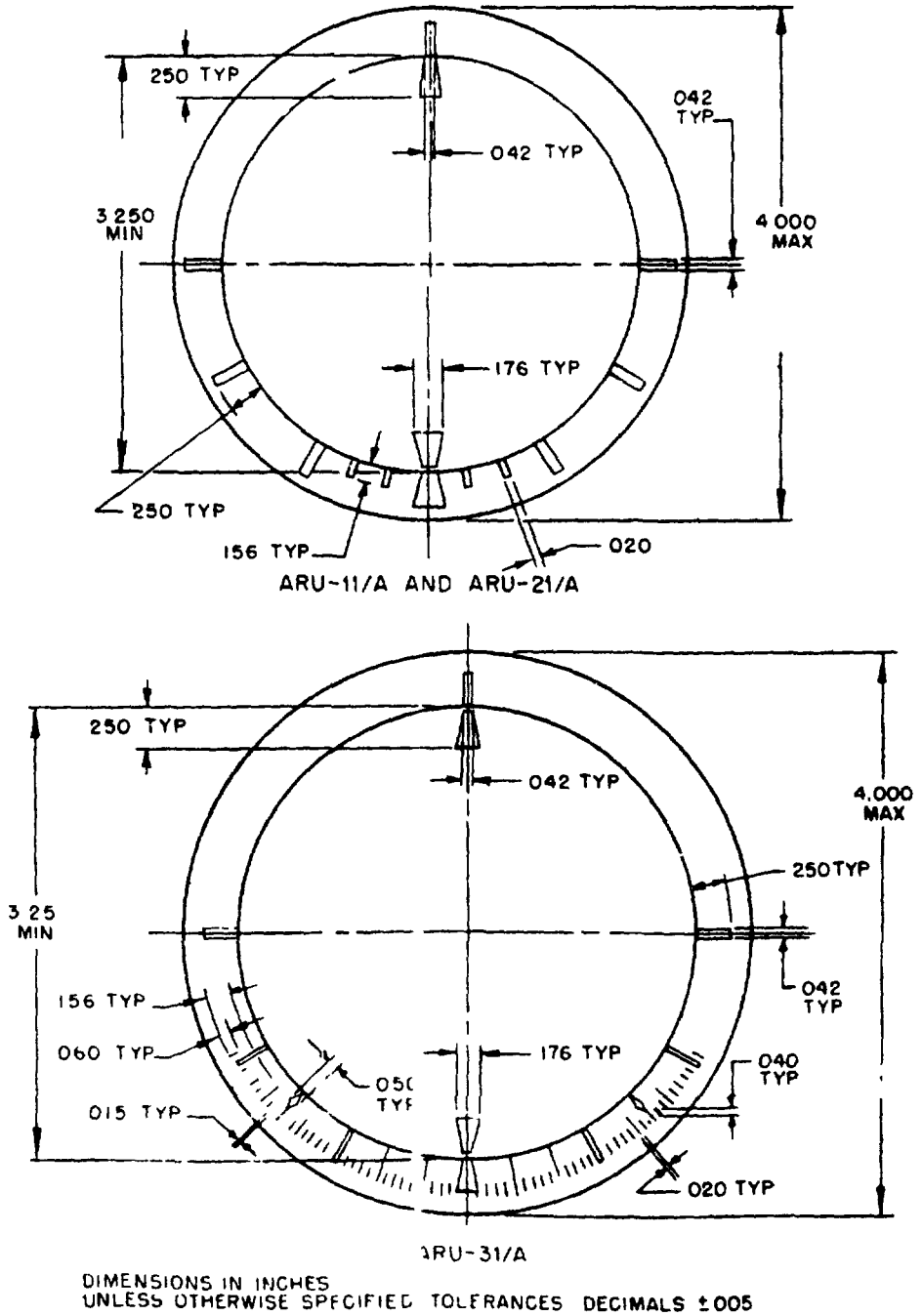


FIGURE 3. Bank Angle and Bank Indexes

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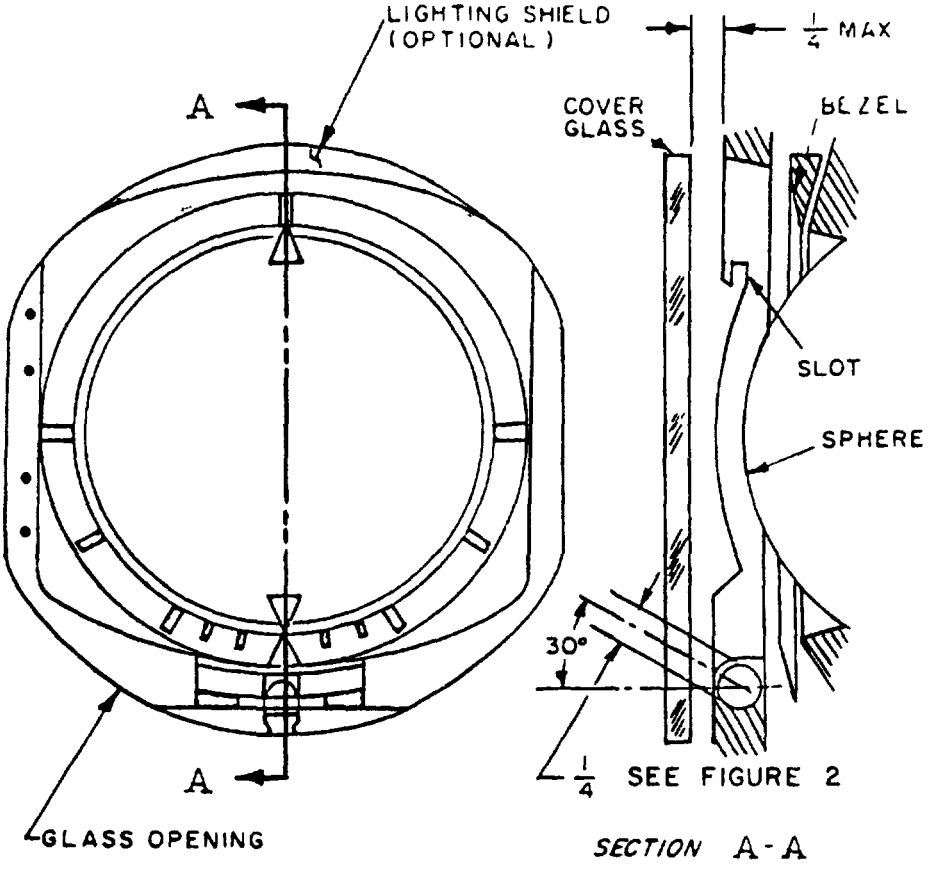
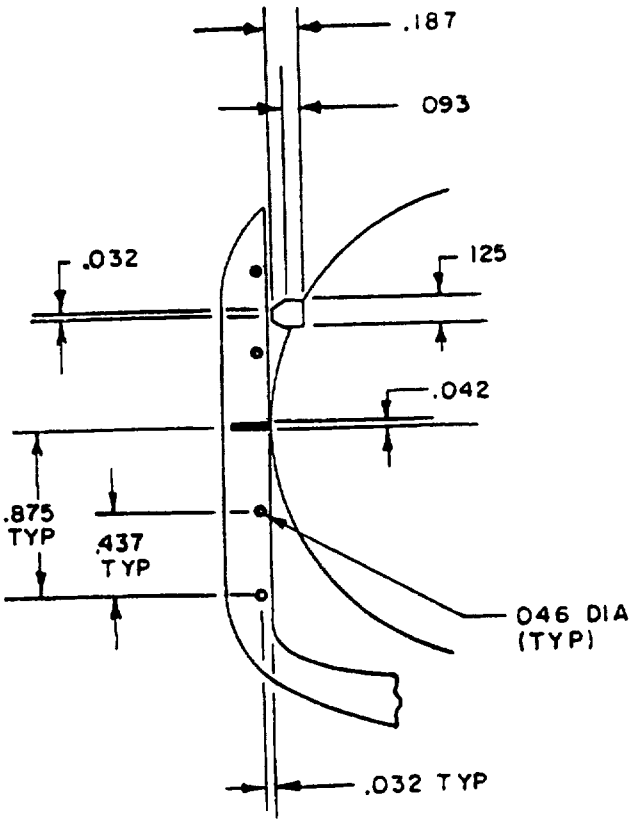


FIGURE 4. Bezel Layout

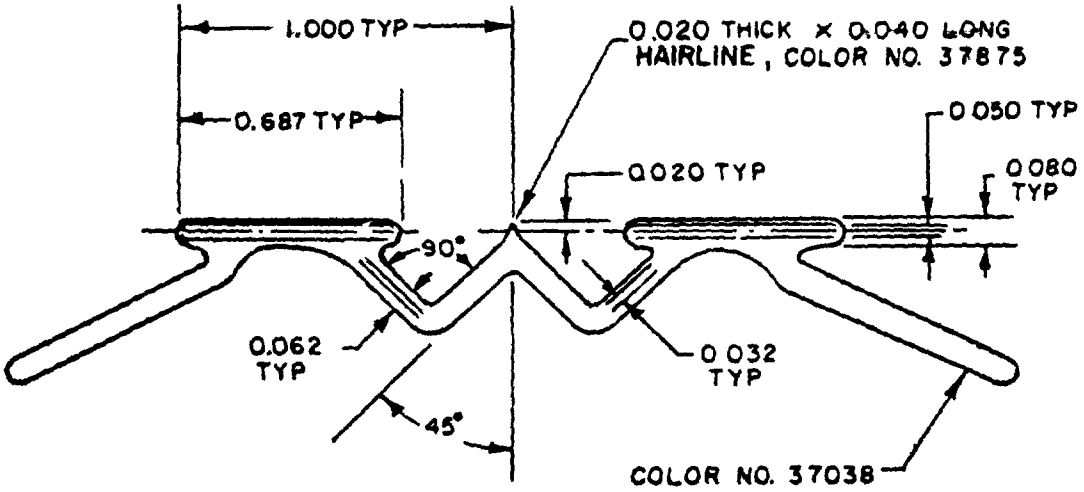
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DIMENSIONS IN INCHES.
UNLESS OTHERWISE SPECIFIED,
TOLERANCES ±.005

FIGURE 5. Dial Mask

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DIMENSIONS IN INCHES
TOLERANCES:
DECIMALS ± 0.005
ANGLES ± 1°

FIGURE 6. Miniature Airplane

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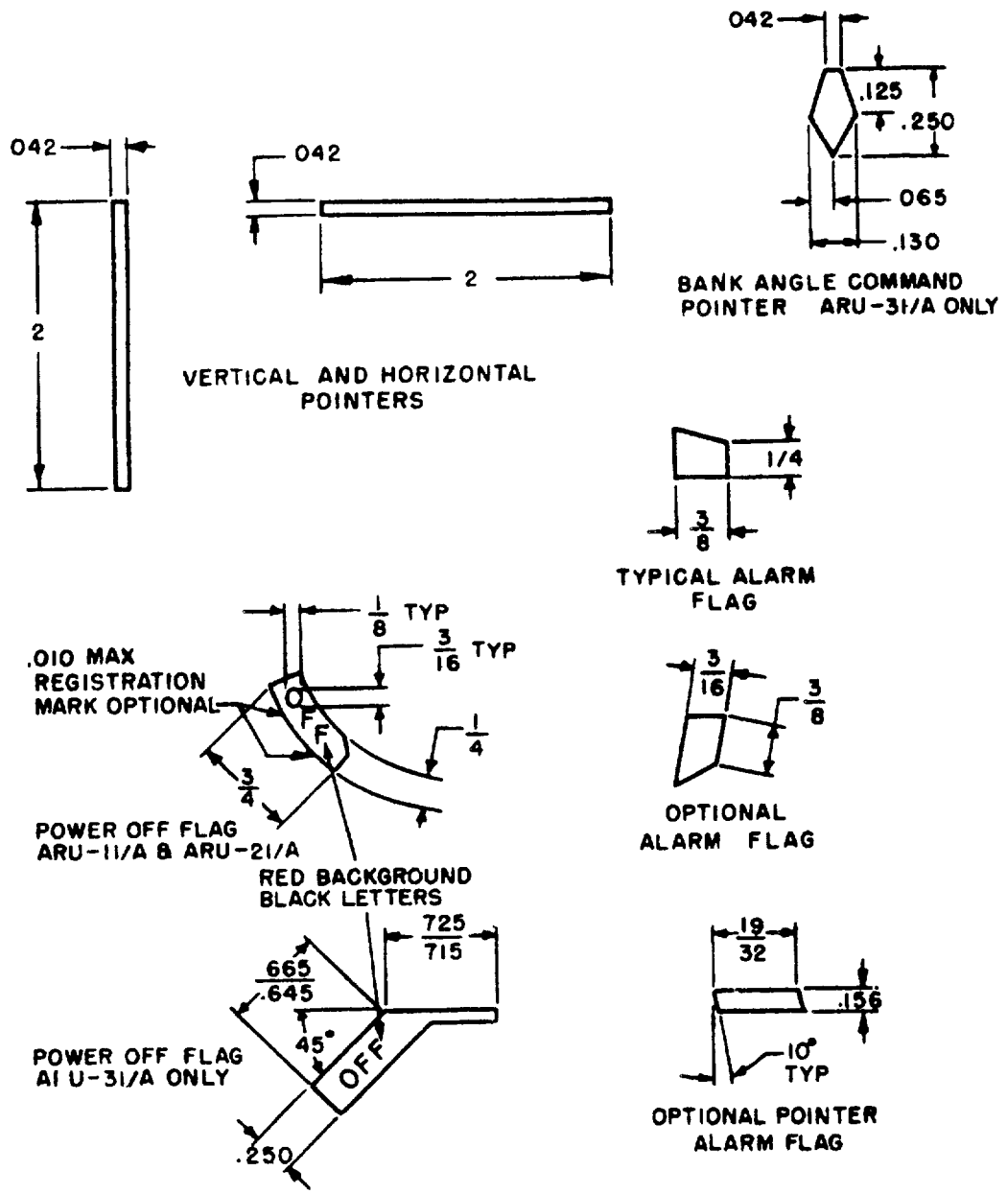


FIGURE 7. Flags and Pointers

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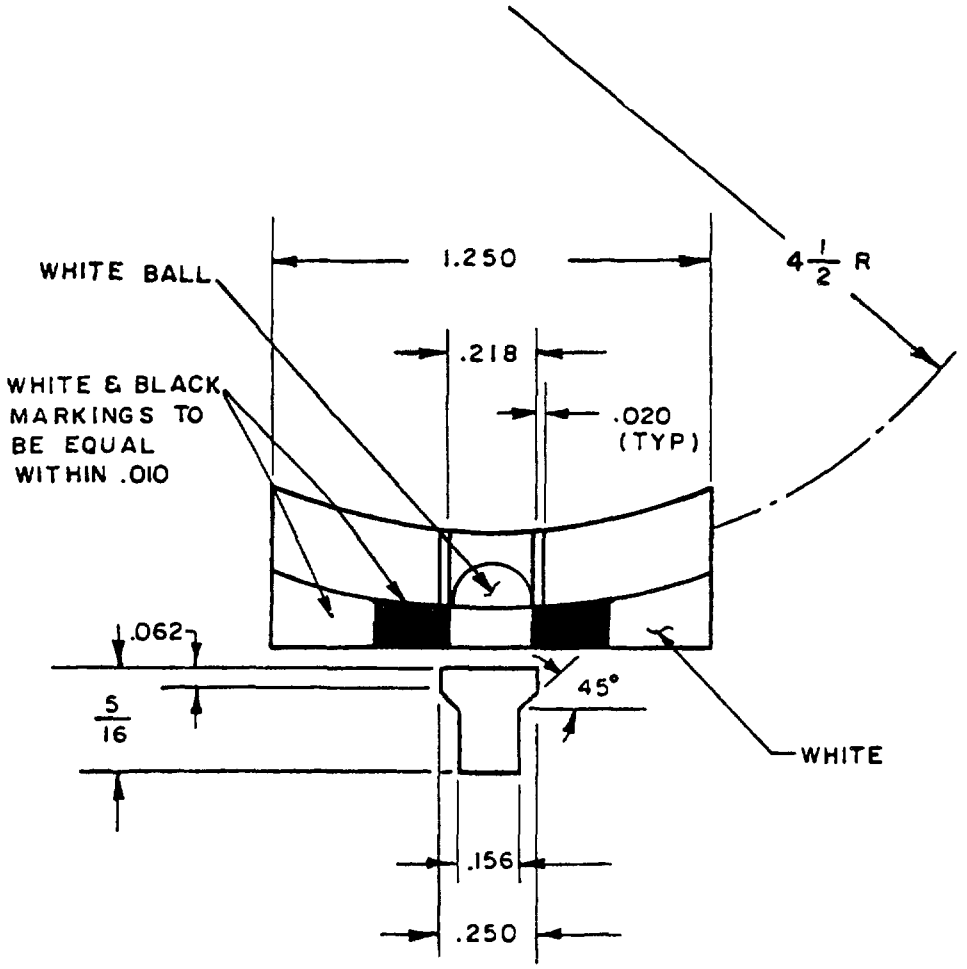


FIGURE 8. Slip Indicator and Rate-of-Turn

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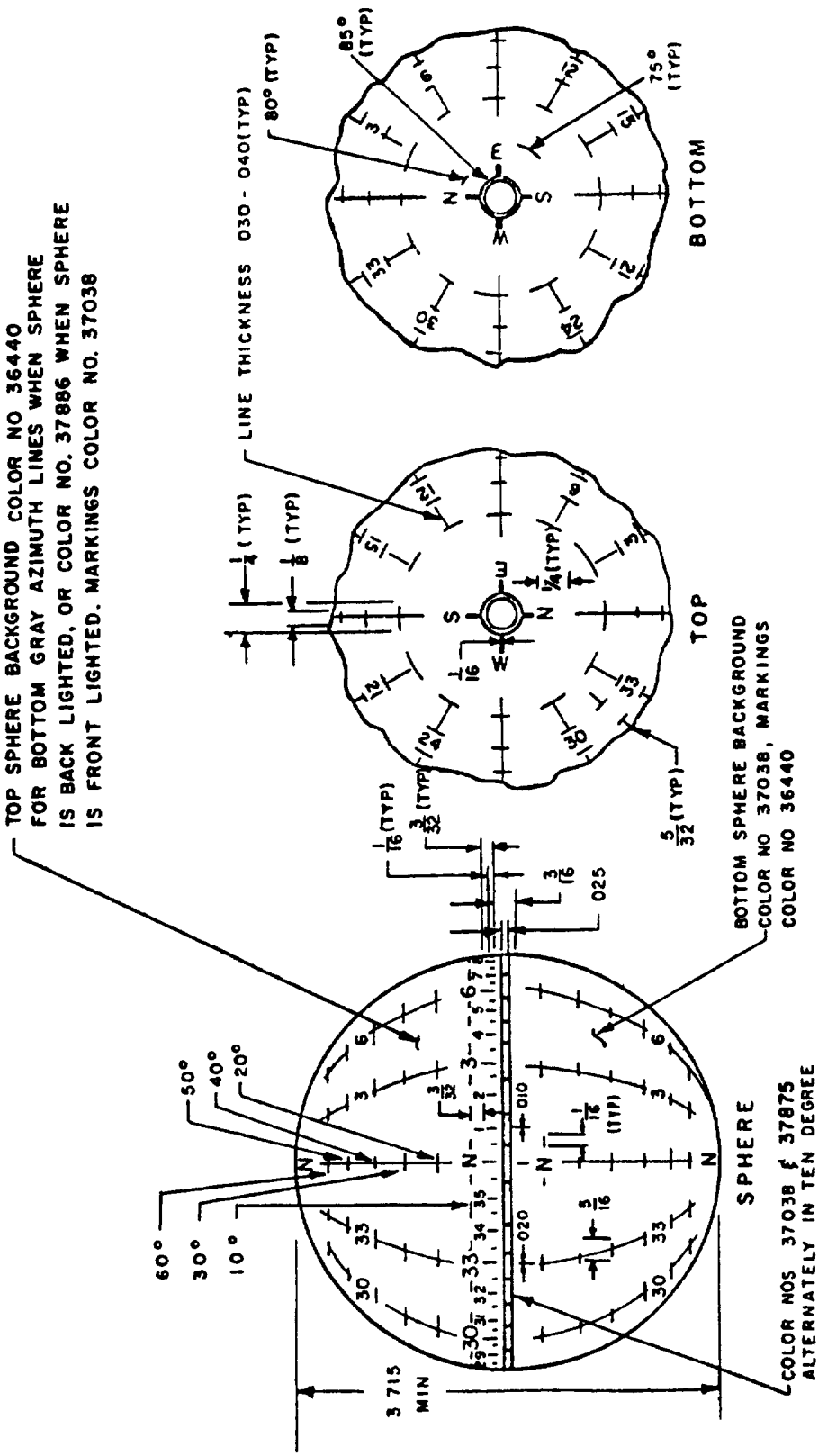
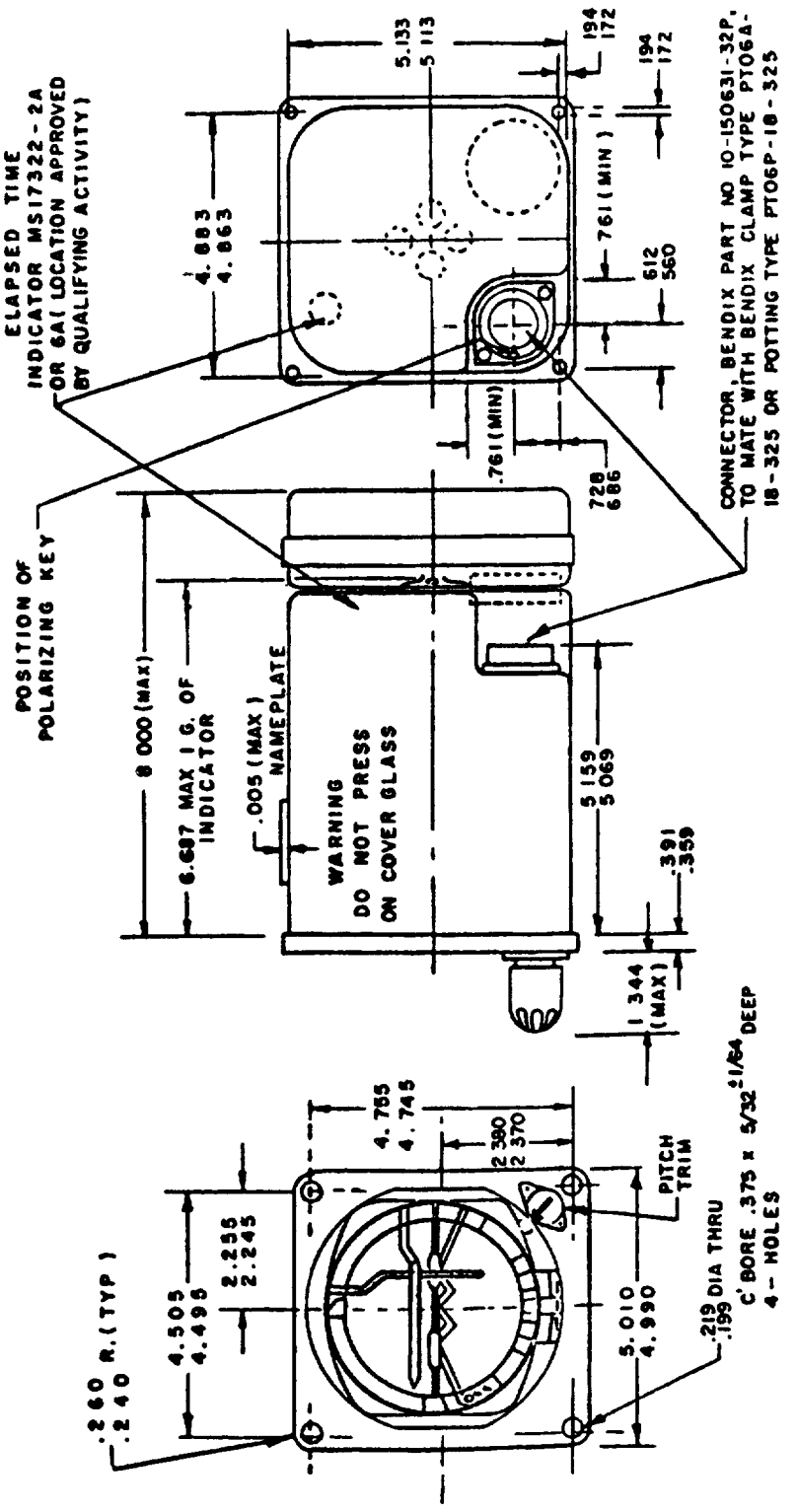


FIGURE 9. Spheroid Markings

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FIGURE 10. Indicator, Attitude-Director

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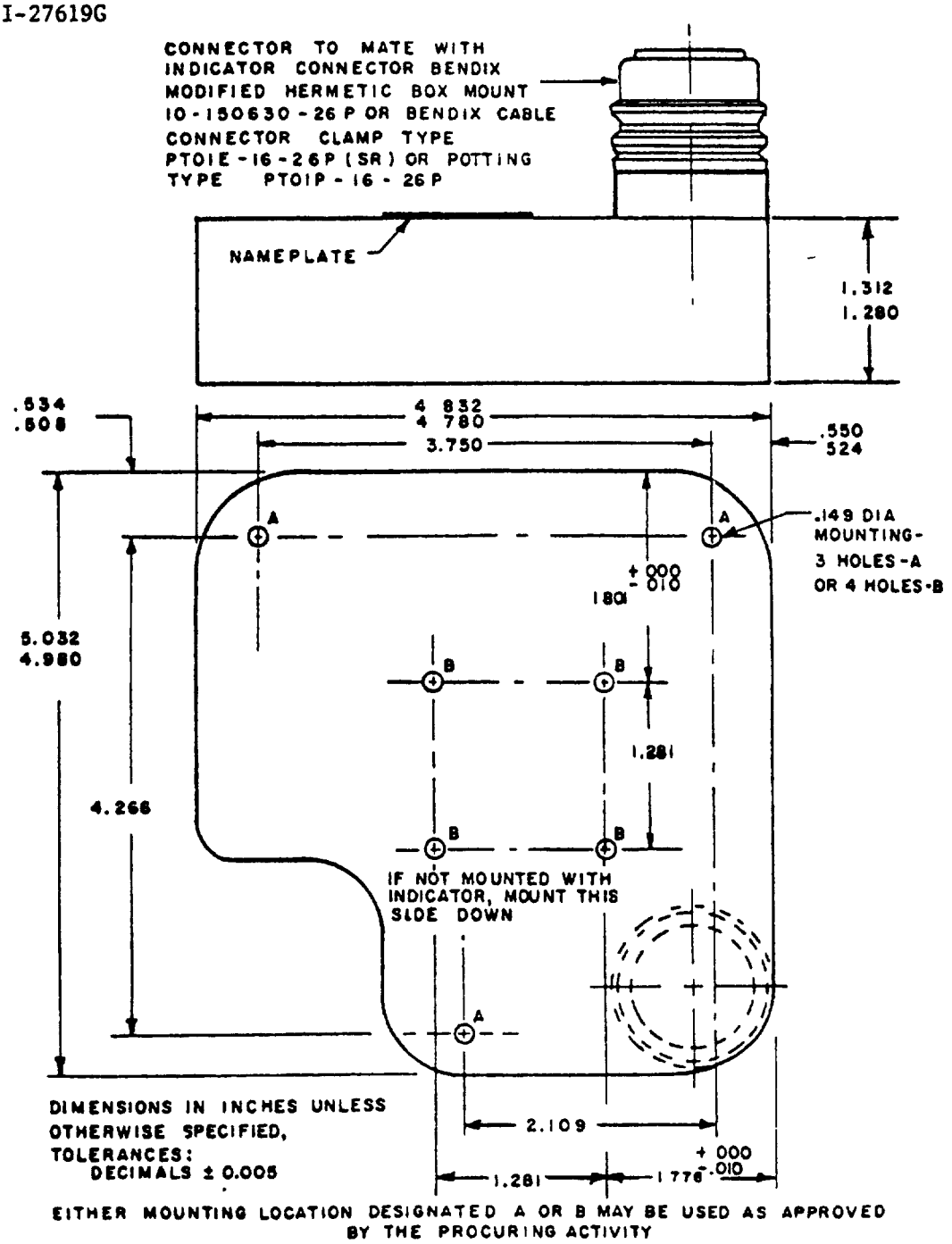


FIGURE 11. Amplifier Envelope

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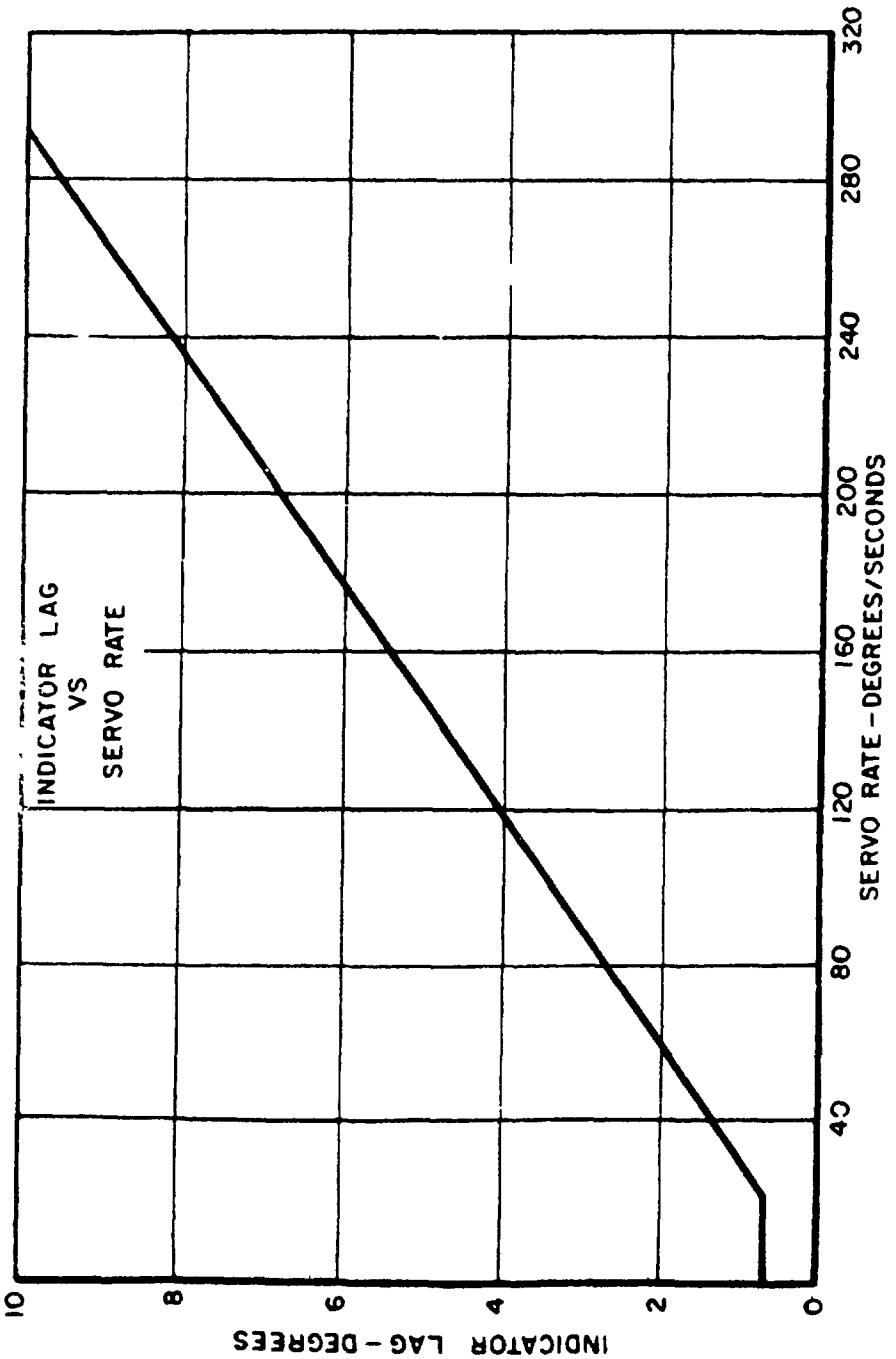
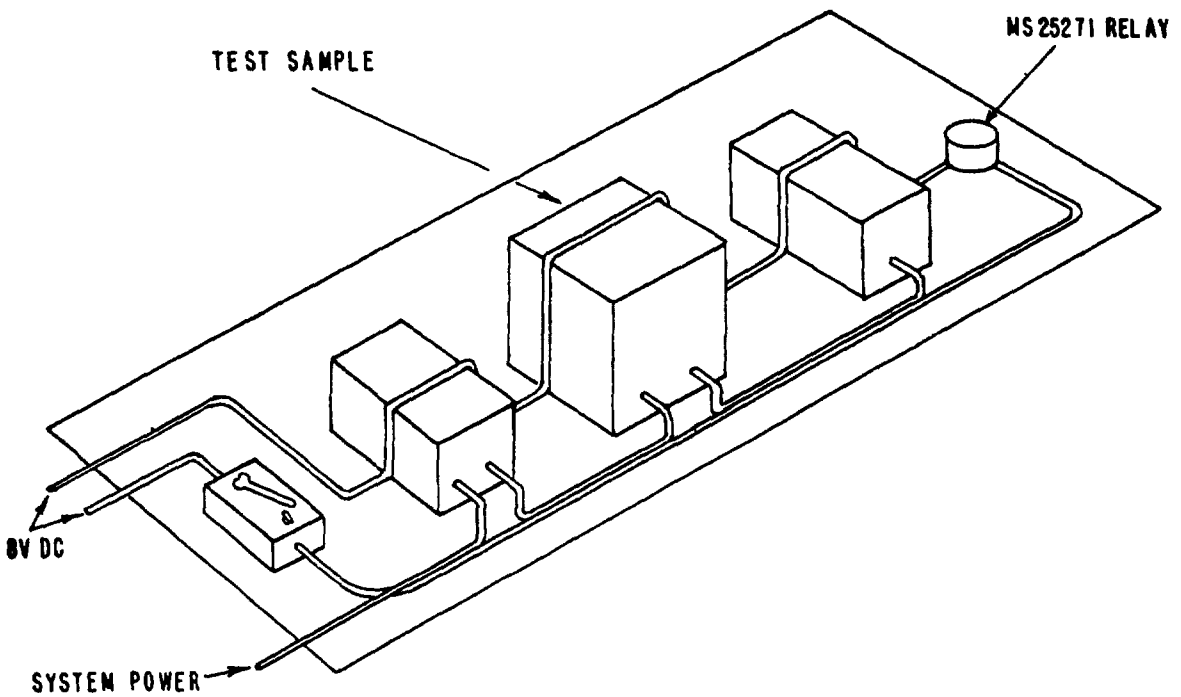


FIGURE 12. Indicator Lag

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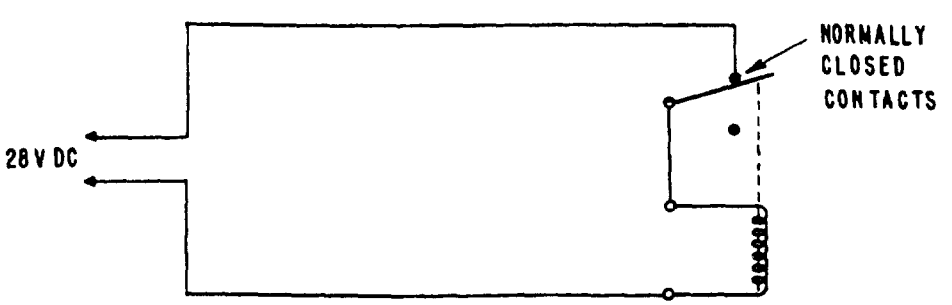


FIGURE 13. Relay Wiring Details

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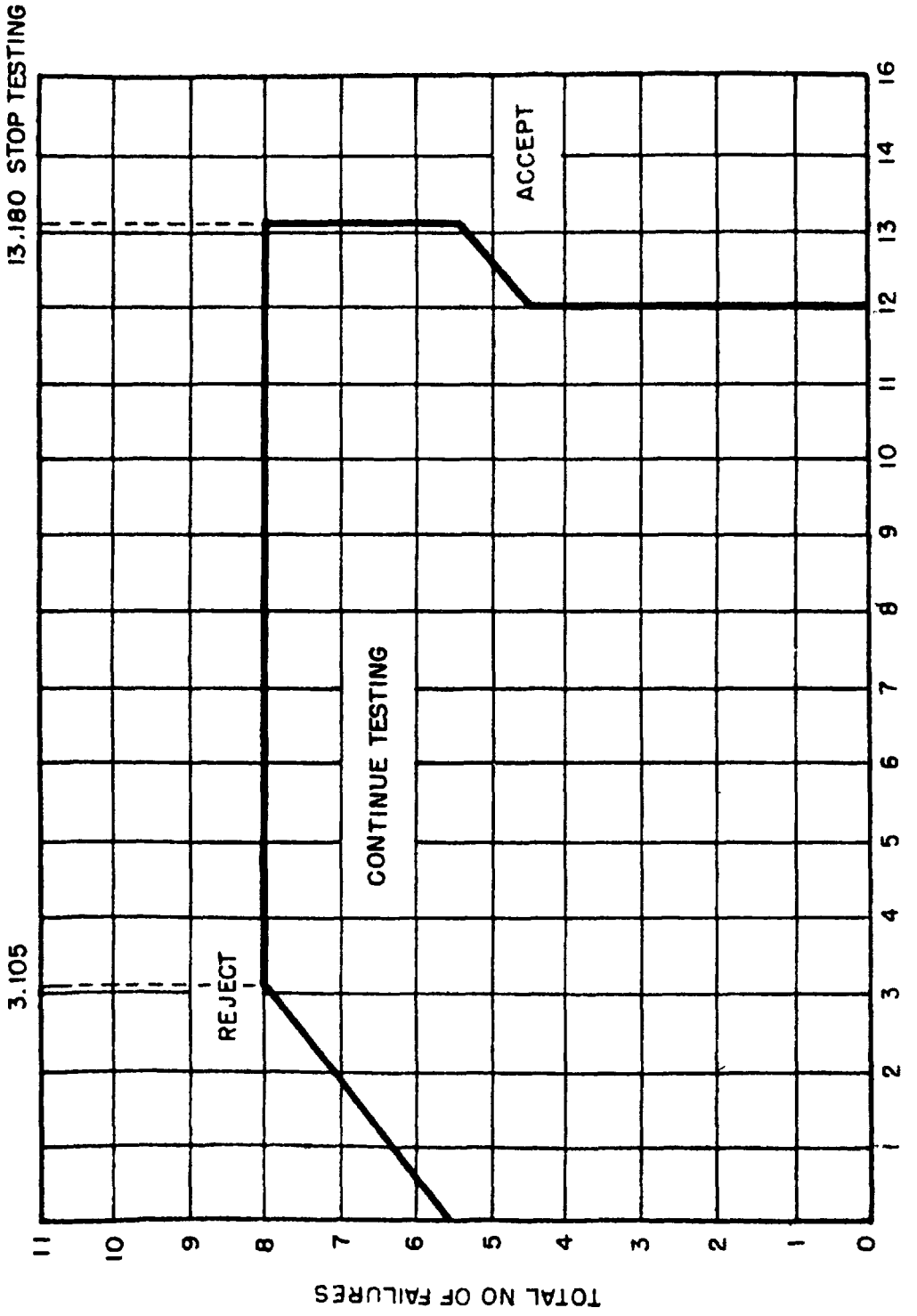


FIGURE 14. Qualification Reliability Accept/Reject Criteria

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